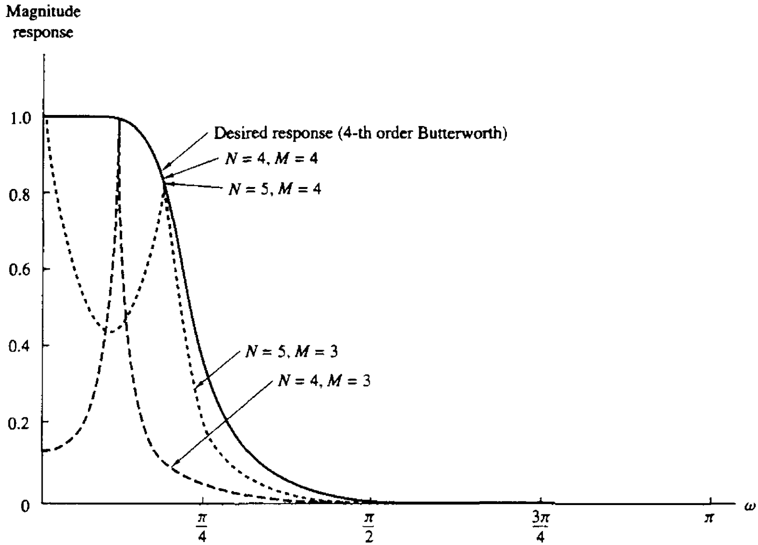


Figure 8.49 Filter designs based on Padè approximation (Example 8.5.2).

In Fig. 8.49 we plot the frequency response of the filter obtained by the Padé approximation method. We have considered four cases: $M = 3$, $N = 5$; $M = 3$, $N = 4$; $M = 4$, $N = 4$; $M = 4$, $N = 5$. We observe that when $M = 3$, the resulting frequency response is a relatively poor approximation to the desired response. However, an increase in the number of poles from $N = 4$ to $N = 5$ appears to compensate in part for the lack of the one zero. When M is increased from three to four, we obtain a perfect match with the desired Butterworth filter not only for $N = 4$ but for $N = 5$, and, in fact, for larger values of N.

Example 8.5.3

A three-pole and three-zero type II lowpass Chebyshev digital filter has the system function

$$H_d(z) = \frac{0.3060(1 + z^{-1})(0.2652 - 0.09z^{-1} + 0.2652z^{-2})}{(1 - 0.3880z^{-1})(1 - 1.1318z^{-1} + 0.5387z^{-2})}$$

Its unit sample response is illustrated in Fig. 8.50. Use the Padé approximation method to approximate $H_d(z)$.

Solution By following the same procedure as in Example 8.5.2, we determined the Padé approximation of $H_d(z)$ based on the selection of $M = 2$, $N = 3$; $M = 2$, $N = 4$; $M = 3$, $N = 3$; $M = 3$, $N = 4$. The frequency responses of the resulting designs are illustrated in Fig. 8.51.

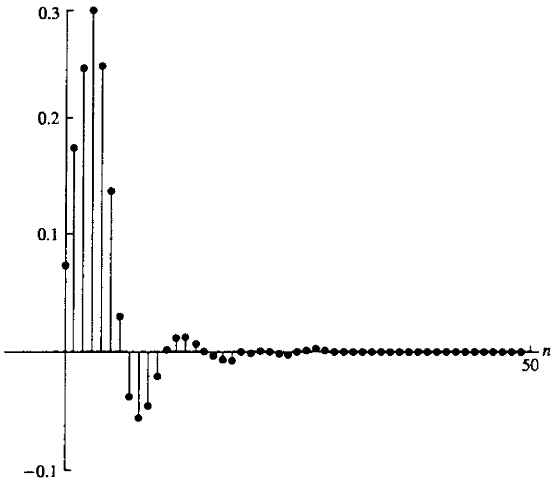


Figure 8.50 Impulse response $h_d(n)$ of type II Chebyshev digital filter given in Example 8.5.3.

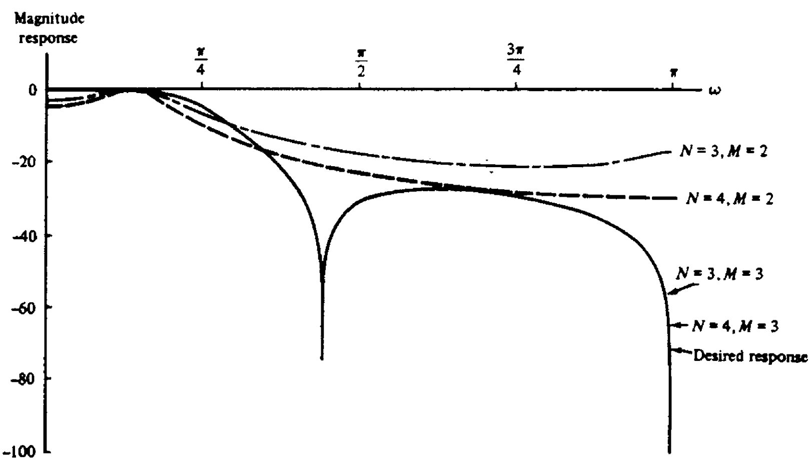


Figure 8.51 Filter designs based on Padé approximation method (Example 8.5.3).

As in Example 8.5.2, we note that when we underestimate the number of zeros we obtain a relatively poor design, as evidenced by the two cases in which $M = 2$. On the other hand, if $M = 3$, we obtain a perfect match for $N = 3$ and $N = 4$.

These two examples suggest that an effective approach in using the Padé approximation is to try different values of M and N until the frequency responses of the resulting filters converge to the desired frequency response within some small, acceptable approximation error. However, in practice, this approach appears to be cumbersome.

8.5.2 Least-Squares Design Methods

Again, let us assume that $h_d(n)$ is specified for $n \geq 0$. We begin with the simple case in which the digital filter to be designed contains only poles, that is,

$$H(z) = \frac{b_0}{1 + \sum_{k=1}^{N} a_k z^{-k}} \tag{8.5.7}$$

Now, consider the cascade connection of the desired filter $H_d(z)$ with the reciprocal, all-zero filter $1/H(z)$, as illustrated in Fig. 8.52. Now suppose that the cascade configuration in Fig. 8.52 is excited by the unit sample sequence $\delta(n)$. Thus the input to the inverse system $1/H(z)$ is $h_d(n)$ and the output is $y(n)$. Ideally, $y_d(n) = \delta(n)$. The actual output is

$$y(n) = \frac{1}{b_0}\left[h_d(n) + \sum_{k=1}^{N} a_k h_d(n-k)\right] \tag{8.5.8}$$

The condition that $y_d(0) \equiv y(0) = 1$ is satisfied by selecting $b_0 = h_d(0)$. For $n > 0$, $y(n)$ represents the error between the desired output $y_d(n) = 0$ and the actual output. Hence the parameters $\{a_k\}$ are selected to minimize the sum of

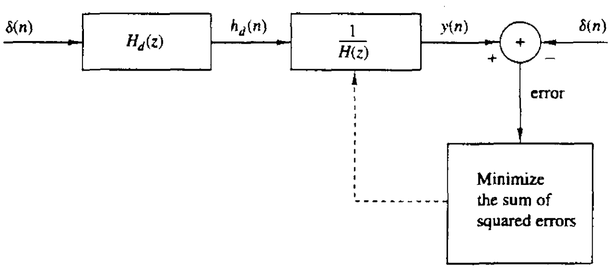


Figure 8.52 Least-squares inverse filter design method.

squares of the error sequence,

$$\begin{aligned}\mathcal{E} &= \sum_{n=1}^{\infty} y^2(n) \\ &= \frac{\sum_{n=1}^{\infty} i \left[h_d(n) + \sum_{k=1}^{N} a_k h_d(n-k) \right]^2}{h_d^2(0)}\end{aligned} \tag{8.5.9}$$

By differentiating with respect to the parameters $\{a_k\}$, it is easily established that we obtain the set of linear equations of the form

$$\sum_{k=1}^{N} a_k r_{hh}(k, l) = -r_{hh}(l, 0) \qquad l = 1, 2, \ldots, N \tag{8.5.10}$$

where, by definition,

$$\begin{aligned}r_{hh}(k, l) &= \sum_{n=1}^{\infty} h_d(n-k) h_d(n-l) \\ &= \sum_{n=0}^{\infty} h_d(n) h_d(n+k-l) = r_{hh}(k-l)\end{aligned} \tag{8.5.11}$$

The solution of (8.5.10) yields the desired parameters for the inverse system $1/H(z)$. Thus we obtain the coefficients of the all-pole filter.

In a practical design problem, the desired impulse response $h_d(n)$ is specified for a finite set of points, say $0 \le n \le L$, where $L >> N$. In such a case, the correlation sequence $r_{dd}(k)$ can be computed from the finite sequence $h_d(n)$ as

$$\hat{r}_{hh}(k-l) = \sum_{n=0}^{L-|k-l|} h_d(n) h_d(n+k-l) \qquad 0 \le k-l \le N \tag{8.5.12}$$

and these values can be used to solve the set of linear equations in (8.5.10).

The least-squares method can also be used in a pole–zero approximation for $H_d(z)$. If the filter $H(z)$ that approximates $H_d(z)$ has both poles and zeros, its response to the unit impulse $\delta(n)$ is

$$h(n) = -\sum_{k=1}^{N} a_k h(n-k) + \sum_{k=0}^{M} b_k \delta(n-k) \qquad n \ge 0 \tag{8.5.13}$$

or, equivalently,

$$h(n) = -\sum_{k=1}^{N} a_k h(n-k) + b_n \qquad 0 \le n \le M \tag{8.5.14}$$

For $n > M$, (8.5.13) reduces to

$$h(n) = -\sum_{k=1}^{N} a_k h(n-k) \qquad n > M \tag{8.5.15}$$

Clearly, if $H_d(z)$ is a pole–zero filter, its response to $\delta(n)$ would satisfy the same equations (8.5.13) through (8.5.15). In general, however, it does not. Nevertheless, we can use the desired response $h_d(n)$ for $n > M$ to construct an estimate of $h_d(n)$, according to (8.5.15). That is,

$$\hat{h}_d(n) = -\sum_{k=1}^{N} a_k h_d(n-k) \tag{8.5.16}$$

Then we can select the filter parameters $\{a_k\}$ to minimize the sum of squared errors between the desired response $h_d(n)$ and the estimate $\hat{h}_d(n)$ for $n > M$. Thus we have

$$\begin{aligned} \mathcal{E}_1 &= \sum_{n=M+1}^{\infty} [h_d(n) - \hat{h}_d(n)]^2 \\ &= \sum_{n=M+1}^{\infty} \left[h_d(n) + \sum_{k=1}^{N} a_k h_d(n-k) \right]^2 \end{aligned} \tag{8.5.17}$$

The minimization of $\mathcal{E}_1$, with respect to the pole parameters $\{a_k\}$, leads to the set of linear equations

$$\sum_{l=1}^{N} a_l r_{hh}(k,l) = -r_{hh}(k,0) \qquad k = 1, 2, \ldots, N \tag{8.5.18}$$

where $r_{hh}(k,l)$ is now defined as

$$r_{hh}(k,l) = \sum_{n=M+1}^{\infty} h_d(n-k) h_d(n-l) \tag{8.5.19}$$

Thus these linear equations yield the filter parameters $\{a_k\}$. Note that these equations reduce to the all-pole filter approximation when M is set to zero.

The parameters $\{b_k\}$ that determine the zeros of the filter can be obtained simply from (8.5.14), where $h(n) = h_d(n)$, by substitution of the values $\{\hat{a}_k\}$ obtained by solving (8.5.18). Thus

$$b_n = h_d(n) + \sum_{k=1}^{N} \hat{a}_k h_d(n-k) \qquad 0 \le n \le M \tag{8.5.20}$$

Therefore, the parameters $\{\hat{a}_k\}$ that determine the poles are obtained by the method of least squares while the parameters $\{b_k\}$ that determine the zeros are obtained by the the Padé approximation method. The foregoing approach for determining the poles and zeros of $H(z)$ is sometimes called *Prony's method.*

The least-squares method provides good estimates for the pole parameters $\{a_k\}$. However, Prony's method may not be as effective in estimating the parameters $\{b_k\}$, primarily because the computation in (8.5.20) is not based on the least-squares method.

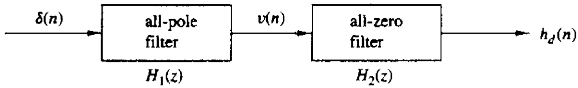


Figure 8.53 Least-squares method for determining the poles and zeros of a filter.

An alternative method in which both sets of parameters $\{a_k\}$ and $\{b_k\}$ are determined by application of the least-squares method has been proposed by Shanks (1967). In Shanks' method, the parameters $\{a_k\}$ are computed on the basis of the least-squares criterion, according to (8.5.18), as indicated above. This yields the estimates $\{\hat{a}_k\}$, which allow us to synthesize the all-pole filter.

$$H_1(z) = \frac{1}{1 + \sum_{k=1}^{N} \hat{a}_k z^{-k}} \tag{8.5.21}$$

The response of this filter to the impulse $\delta(n)$ is

$$v(n) = -\sum_{k=1}^{N} \hat{a}_k v(n-k) + \delta(n) \qquad n \geq 0 \tag{8.5.22}$$

If the sequence $\{v(n)\}$ is used to excite an all-zero filter with system function

$$H_2(z) = \sum_{k=0}^{M} b_k z^{-k} \tag{8.5.23}$$

as illustrated in Fig. 8.53, its response is

$$\hat{h}_d(n) = \sum_{k=0}^{M} b_k v(n-k) \tag{8.5.24}$$

Now we can define an error sequence $e(n)$ as

$$\begin{aligned} e(n) &= h_d(n) - \hat{h}_d(n) \\ &= h_d(n) - \sum_{k=0}^{M} b_k v(n-k) \end{aligned} \tag{8.5.25}$$

and, consequently, the parameters $\{b_k\}$ can also be determined by means of the least-squares criterion, namely, from the minimization of

$$\mathcal{E}_2 = \sum_{n=0}^{\infty} \left[h_d(n) - \sum_{k=0}^{M} b_k v(n-k) \right]^2 \tag{8.5.26}$$

Thus we obtain a set of linear equations for the parameters $\{b_k\}$, in the form

$$\sum_{k=0}^{M} b_k r_{vv}(k, l) = r_{hv}(l) \qquad l = 0, 1, \ldots, M \tag{8.5.27}$$

where, by definition,

$$r_{vv}(k,l) = \sum_{n=0}^{\infty} v(n-k)v(n-l) \tag{8.5.28}$$

$$r_{hv}(k) = \sum_{n=0}^{\infty} h_d(n)v(n-k) \tag{8.5.29}$$

Example 8.5.4

Approximate the fourth-order Butterworth filter given in Example 8.5.2 by means of an all-pole filter using the least-squares inverse design method.

Solution From the desired impulse response $h_d(n)$, which is illustrated in Fig. 8.48, we computed the autocorrelation sequence $r_{hh}(k,l) = r_{hh}(k-l)$ and solved to set of linear equations in (8.5.10) to obtain the filter coefficients. The results of this computation are given in Table 8.14 for $N = 3, 4, 5, 10$, and 15. In Table 8.15 we list the poles of the filter designs for $N = 3, 4$, and 5 along with the actual poles of the fourth-order Butterworth filter. We note that the poles obtained from the designs are far from the actual poles of the desired filter.

TABLE 8.14 ESTIMATES OF FILTER COEFFICIENTS $\{a_k\}$ IN LEAST-SQUARES INVERSE FILTER DESIGN METHOD

$N = 3$	$N = 15$
$a_1 = 0.254295E+01$	$a_1 = 2.993620$
$a_2 = -0.241800E+01$	$a_2 = -1.143053$
$a_3 = 0.853829E+00$	$a_3 = -12.132861$
$N = 4$	$a_4 = 39.663433$
$a_1 = 0.319047E+01$	$a_5 = -75.749001$
$a_2 = -0.425176E+01$	$a_6 = 109.247757$
$a_3 = 0.278234E+01$	$a_7 = -129.513794$
$a_4 = 0.758375E+00$	$a_8 = 131.026794$
$N = 5$	$a_9 = -114.905266$
$a_1 = 0.368733E+01$	$a_{10} = 87.449211$
$a_2 = -0.607422E+01$	$a_{11} = -57.031906$
$a_3 = 0.556726E+01$	$a_{12} = 30.915134$
$a_4 = -0.284813E+01$	$a_{13} = -13.124536$
$a_5 = 0.654996E+00$	$a_{14} = 3.879295$
$N = 10$	$a_{15} = -0.597313$
$a_1 = 5.008451$	
$a_2 = -12.660761$	
$a_3 = 21.557365$	
$a_4 = -27.804110$	
$a_5 = 28.683949$	
$a_6 = -24.058558$	
$a_7 = 16.156847$	
$a_8 = -8.247148$	
$a_9 = 2.854789$	
$a_{10} = -0.502956$	

TABLE 8.15 ESTIMATES OF POLE POSITIONS IN LEAST-SQUARES INVERSE FILTER DESIGN METHOD (EXAMPLE 8.5.4)

Number of poles	Pole positions
$N = 3$	0.9305
	$0.8062 \pm j0.5172$
$N = 4$	$0.8918 \pm j0.2601$
	$0.7037 \pm j0.6194$
$N = 5$	0.914
	$0.8321 \pm j0.4307$
	$0.5544 \pm j0.7134$
$N = 4$	$0.6603 \pm j0.4435$
Butterworth filter	$0.5241 \pm j0.1457$

The frequency responses of the filter designs are plotted in Fig. 8.54. We note that when N is small, the approximation to the desired filter is poor. As N is increased to $N = 10$ and $N = 15$, the approximation improves significally. However, even for $N = 15$, there are large ripples in the passband of the filter response. It is apparent that this method, which is based on an all-pole approximation, does not provide good approximations to filters that contain zeros.

Example 8.5.5

Approximate the type II Chebyshev lowpass filter given in Example 8.5.3 by means of the three least-squares methods described above.

Solution The results of the filter designs obtained by means of the least-squares inverse method, Prony's method and Shanks' method, are illustrated in Fig. 8.55. The filter parameters obtained from these design methods are listed in Table 8.16.

The frequency response characteristics in Fig. 8.55 illustrate that the least-squares inverse (all-pole) design method yields poor designs when the filter contains zeros. On the other hand, both Prony's method and Shanks' method yield very good designs when the number of poles and zeros equals or exceeds the number of poles and zeros in the actual filter. Thus the inclusion of zeros in the approximation has a significant effect in the resulting filter design.

8.5.3 FIR Least-Squares Inverse (Wiener) Filters

In the preceding section we described the use of the least-squares error criterion in the design of pole–zero filters. In this section we use a similar approach to determine a least-squares FIR inverse filter to a desired filter.

The inverse to a linear time-invariant system with impulse response $h(n)$ and system function $H(z)$ is defined as the system whose impulse response $h_I(n)$ and

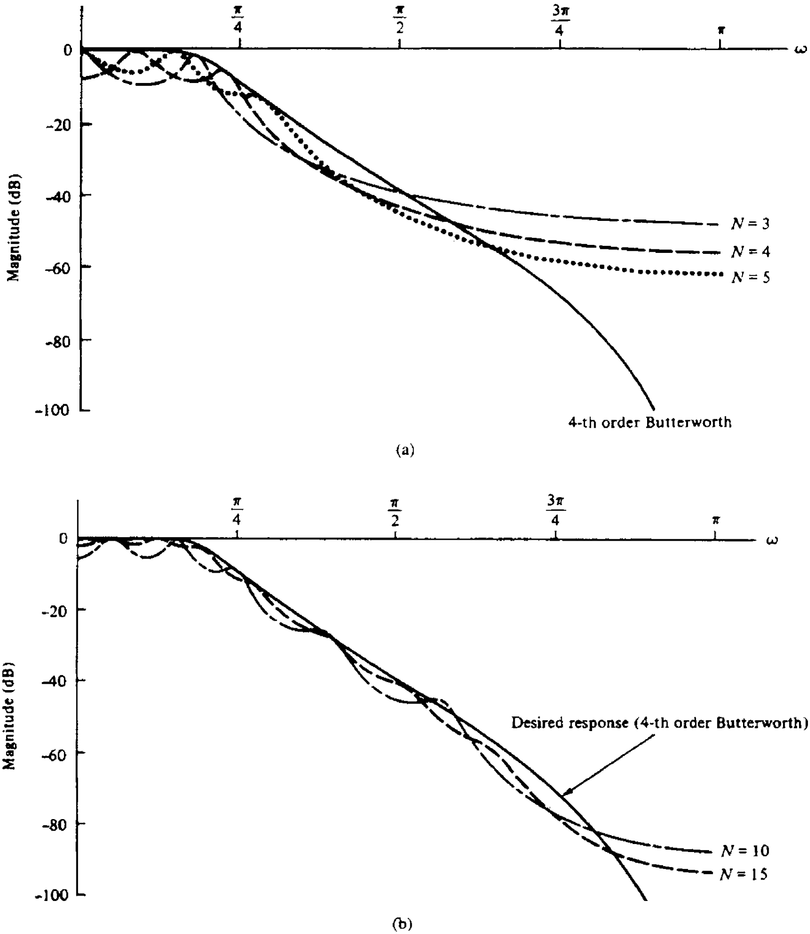


Figure 8.54 Magnitude responses for filter designs based on the least-squares inverse filter method.

system function $H_I(z)$, satisfy the respective equations.

$$h(n) * h_I(n) = \delta(n) \tag{8.5.30}$$

$$H(z)H_I(z) = 1 \tag{8.5.31}$$

In general, $H_I(z)$ is IIR, unless $H(z)$ is an all-pole system, in which case $H_I(z)$ is FIR.

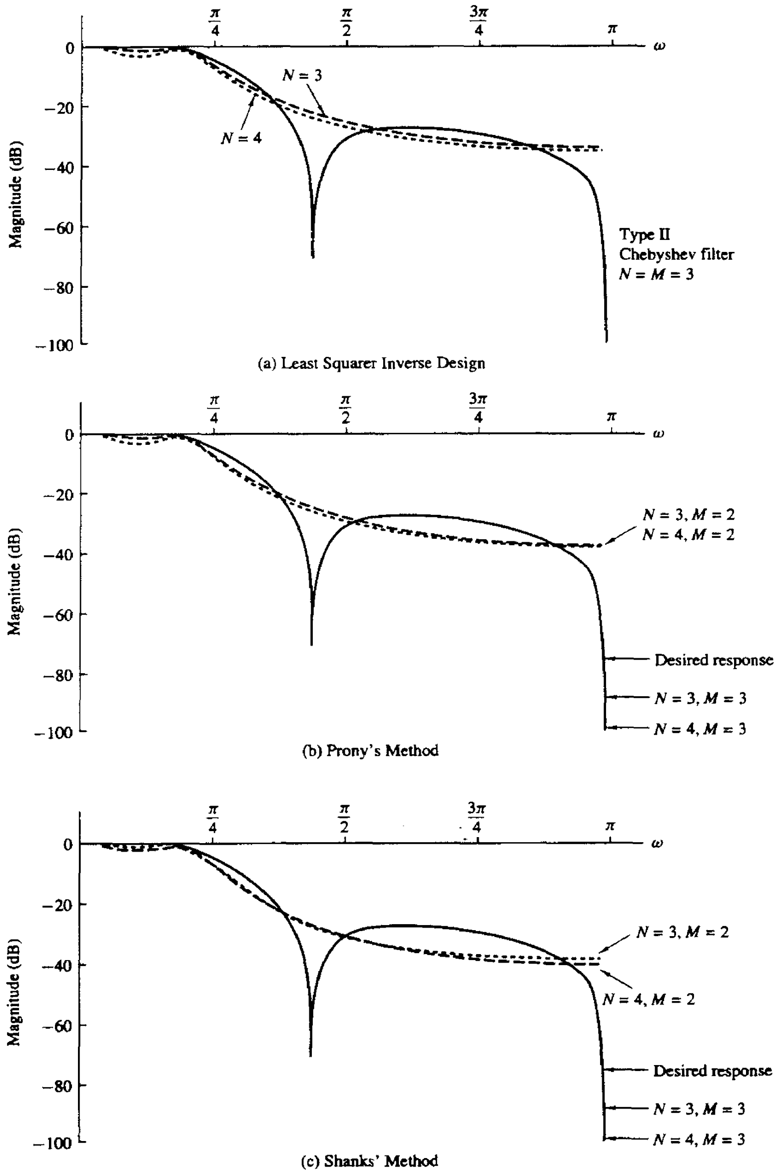


Figure 8.55 Filter designs based on least-squares methods (Example 8.5.5): (a) least-squares design; (b) Prony's method; (c) Shank's method.

TABLE 8.16 POLE-ZERO LOCATIONS FOR FILTER DESIGNS IN EXAMPLE 8.5.5

Chebyshev Filter:
Zeros: $-1, 0.1738311 \pm j0.9847755$
Poles: $0.3880, 0.5659 \pm j0.467394$

Filter Order	Poles in Least-Squares Inverse
$N = 3$	0.8522
	$0.6544 \pm j0.6224$
$N = 4$	$0.7959 \pm j0.3248$
	$0.4726 \pm j0.7142$

Filter Order	Prony's Method		Shanks' Method	
	Poles	Zeros	Poles	Zeros
$N = 3$	0.5332		0.5348	
$M = 2$	$0.6659 \pm j0.4322$	$-0.1497 \pm j0.4925$	$0.6646 \pm j0.4306$	$-0.2437 \pm j0.5918$
$N = 4$	0.7092		0.7116	
	−0.2919		−0.2921	
$M = 2$	$0.6793 \pm j0.4863$	$-0.1982 \pm j0.37$	$0.6783 \pm j0.4855$	$0.306 \pm j0.4482$
$N = 3$	0.3881	−1	0.3881	−1
$M = 3$	$0.5659 \pm j0.4671$	$0.1736 \pm j0.9847$	$0.5659 \pm j0.4671$	$0.1738 \pm j0.9848$
$N = 4$	−0.00014	−1	−0.00014	−1
	0.388		0.388	
$M = 3$	$0.5661 \pm j0.4672$	$0.1738 \pm j0.9848$	$0.566 \pm j0.4671$	$0.1738 \pm j0.9848$

In many practical applications, it is desirable to restrict the inverse filter to be FIR. Obviously, one simple method is to truncate $h_I(n)$. In so doing, we incur a total squared approximation error equal to

$$\mathcal{E}_t = \sum_{n=M+1}^{\infty} h_I^2(n) \tag{8.5.32}$$

where $M + 1$ is the length of the truncated filter and $\mathcal{E}_t$ represents the energy in the tail of the impulse response $h_I(n)$.

Alternatively, we can use the least-squares error criterion to optimize the $M + 1$ coefficients of the FIR filter. First, let $d(n)$ denote the *desired output sequence* of the FIR filter of length $M + 1$ and let $h(n)$ be the input sequence. Then, if $y(n)$ is the output sequence of the filter, as illustrated in Fig. 8.56, the error sequence between the desired output and the actual output is

$$e(n) = d(n) - \sum_{k=0}^{M} b_k h(n-k) \tag{8.5.33}$$

where the $\{b_k\}$ are the FIR filter coefficients.

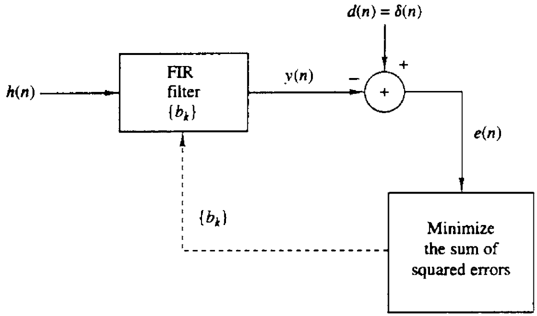


Figure 8.56 Least-squares FIR inverse filter.

The sum of squares of the error sequence is

$$\mathcal{E} = \sum_{n=0}^{\infty} \left[d(n) - \sum_{k=0}^{M} b_k h(n-k) \right]^2 \tag{8.5.34}$$

When $\mathcal{E}$ is minimized with respect to the filter coefficients, we obtain the set of linear equations

$$\sum_{k=0}^{M} b_k r_{hh}(k-l) = r_{dh}(l) \qquad l = 0, 1, \ldots, M \tag{8.5.35}$$

where $r_{hh}(l)$ is the autocorrelation of $h(n)$, defined as

$$r_{hh}(l) = \sum_{n=0}^{\infty} h(n)h(n-l) \tag{8.5.36}$$

and $r_{dh}(n)$ is the crosscorrelation between the desired output $d(n)$ and the input sequence $h(n)$, defined as

$$r_{dh}(l) = \sum_{n=0}^{\infty} d(n)h(n-l) \tag{8.5.37}$$

The optimum, in the least-squares sense, FIR filter that satisfies the linear equations in (8.5.35) is called the *Wiener filter*, after the famous mathematician Norbert Wiener, who introduced optimum least-squares filtering methods in engineering [see book by Wiener (1949)].

If the optimum least-squares FIR filter is to be an approximate inverse filter, the desired response is

$$d(n) = \delta(n) \tag{8.5.38}$$

The crosscorrelation between $d(n)$ and $h(n)$ reduces to

$$r_{dh}(l) = \begin{cases} h(0) & l = 0 \\ 0 & \text{otherwise} \end{cases} \tag{8.5.39}$$

Therefore, the coefficients of the least-squares FIR filter are obtained from the solution of the linear equations in (8.5.35), which can be expressed in matrix form as

$$\begin{bmatrix} r_{hh}(0) & r_{hh}(1) & r_{hh}(2) & \cdots & r_{hh}(M) \\ r_{hh}(1) & r_{hh}(0) & r_{hh}(1) & \cdots & r_{hh}(M-1) \\ \vdots & \vdots & \vdots & & \vdots \\ r_{hh}(M) & r_{hh}(M-1) & \cdots & \cdots & r_{hh}(0) \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_M \end{bmatrix} = \begin{bmatrix} h(0) \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{8.5.40}$$

We observe that the matrix is not only symmetric but it also has the special property that all the elements along any diagonal are equal. Such a matrix is called a Toeplitz matrix and lends itself to efficient inversion by means of an algorithm due to Levinson (1947) and Durbin (1959), which requires a number of computations proportional to M^2 instead of the usual M^3. The Levinson–Durbin algorithm is described in Chapter 11.

The minimum value of the least-squares error obtained with the optimum FIR filter is

$$\begin{aligned} \mathcal{E}_{\min} &= \sum_{n=0}^{\infty} \left[d(n) - \sum_{k=0}^{M} b_k h(n-k) \right] d(n) \\ &= \sum_{n=0}^{\infty} d^2(n) - \sum_{k=0}^{M} b_k r_{dh}(k) \end{aligned} \tag{8.5.41}$$

In the case where the FIR filter is the least-squares inverse filter, $d(n) = \delta(n)$ and $r_{dh}(n) = h(0)\delta(n)$. Therefore,

$$\mathcal{E}_{\min} = 1 - h(0)b_0 \tag{8.5.42}$$

Example 8.5.6

Determine the least-squares FIR inverse filter of length 2 to the system with impulse response

$$h(n) = \begin{cases} 1, & n = 0 \\ -\alpha, & n = 1 \\ 0, & \text{otherwise} \end{cases}$$

where $|\alpha| < 1$. Compare the least-squares solution with the approximate inverse obtained by truncating $h_I(n)$.

Solution Since the system has a system function $H(z) = 1 - \alpha z^{-1}$, the exact inverse is IIR and is given by

$$H_I(z) = \frac{1}{1 - \alpha z^{-1}}$$

or, equivalently,

$$h_I(n) = \alpha^n u(n)$$

If this is truncated after n terms, the residual energy in the tail is

$$\mathcal{E}_t = \sum_{k=n}^{\infty} \alpha^{2k}$$

$$\mathcal{E}_t = \alpha^{2n}(1 + \alpha^2 + \alpha^4 + \cdots)$$

$$\mathcal{E}_t = \frac{\alpha^{2n}}{1-\alpha^2}$$

From (8.5.40) the least-squares FIR filter of length 2 satisfies the equations

$$\begin{bmatrix} 1+\alpha^2 & -\alpha \\ -\alpha & 1+\alpha^2 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

which have the solution

$$b_0 = \frac{1+\alpha^2}{1+\alpha^2+\alpha^4}$$

$$b_1 = \frac{\alpha}{1+\alpha^2+\alpha^4}$$

For purposes of comparison, the truncated inverse filter of length 2 has the coefficients $b_0 = 1$, $b_1 = \alpha$.

The least-squares error is

$$\mathcal{E}_{\min} = \frac{\alpha^4}{1+\alpha^2+\alpha^4}$$

which compares with

$$\mathcal{E}_t = \frac{\alpha^4}{1-\alpha^2}$$

for the truncated approximate inverse. Clearly, $\mathcal{E}_t > \mathcal{E}_{\min}$, so that the least-squares FIR inverse filter is superior.

In this example, the impulse response $h(n)$ of the system was minimum phase. In such a case we selected the desired response to be $d(0) = 1$ and $d(n) = 0$, $n \geq 1$. On the other hand, if the system is nonminimum phase, a delay should be inserted in the desired response in order to obtain a good filter design. The value of the appropriate delay depends on the characteristics of $h(n)$. In any case we can compute the least-squares error filter for different delays and select the filter that produces the smallest error. The following example illustrates the effect of the delay.

Example 8.5.7

Determine the least-squares FIR inverse of length 2 to the system with impulse response

$$h(n) = \begin{cases} -\alpha, & n = 0 \\ 1, & n = 1 \\ 0, & \text{otherwise} \end{cases}$$

where $|\alpha| < 1$.

Solution This is a maximum-phase system. If we select $d(n) = [1 \quad 0]$ we obtain the same solution as in Example 8.5.6, with a minimum least-squares error

$$\begin{aligned}\mathcal{E}_{\min} &= 1 - h(0)b_0 \\ &= 1 + \alpha \frac{1+\alpha^2}{1+\alpha^2+\alpha^4}\end{aligned}$$

If $0 < \alpha < 1$, then $\mathcal{E}_{\min} > 1$, which represents a poor inverse filter. If $-1 < \alpha < 0$, then $\mathcal{E}_{\min} < 1$. In particular, for $\alpha = \frac{1}{2}$, we obtain $\mathcal{E}_{\min} = 1.57$. For $\alpha = -\frac{1}{2}$, $\mathcal{E}_{\min} = 0.81$, which is still a very large value for the squared error.

Now suppose that the desired response is specified as $d(n) = \delta(n-1)$. Then the set of equations for the filter coefficients, obtained from (8.5.35), are the solution to the equations

$$\begin{bmatrix} 1+\alpha^2 & -\alpha \\ -\alpha & 1+\alpha^2 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} h(1) \\ h(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -\alpha \end{bmatrix}$$

The solution of these equations is

$$\begin{aligned} b_0 &= \frac{1}{1+\alpha^2+\alpha^4} \\ b_1 &= \frac{-\alpha^3}{1+\alpha^2+\alpha^4} \end{aligned}$$

The least-squares error, given by (8.5.41), is

$$\begin{aligned}\mathcal{E}_{\min} &= 1 - b_0 r_{dh}(0) - b_1 r_{dh}(1) \\ &= 1 - b_0 h(1) - b_1 h(0) \\ \mathcal{E}_{\min} &= 1 - \frac{1}{1+\alpha^2+\alpha^4} + \frac{\alpha^4}{1+\alpha^2+\alpha^4} \\ \mathcal{E}_{\min} &= 1 - \frac{1-\alpha^4}{1+\alpha^2+\alpha^4}\end{aligned}$$

In particular, suppose that $\alpha = \pm\frac{1}{2}$. Then $\mathcal{E}_{\min} = 0.29$. Consequently, the desired response $d(n) = \delta(n-1)$ results in a significantly better inverse filter. Further improvement is possible by increasing the length of the inverse filter.

In general, when the desired response is specified to contain a delay D, then the crosscorrelation $r_{dh}(l)$, defined in (8.5.37), becomes

$$r_{dh}(l) = h(D-l) \qquad l = 0, 1, \ldots, M$$

The set of linear equations for the coefficients of the least-squares FIR inverse filter given by (8.5.35) reduce to

$$\sum_{k=0}^{M} b_k r_{hh}(k-l) = h(D-l) \qquad l = 0, 1, \ldots, M \tag{8.5.43}$$

Then the expression for the corresponding least-squares error, given in general by

(8.5.41), becomes

$$\mathcal{E}_{\min} = 1 - \sum_{k=0}^{M} b_k h(D-k) \tag{8.5.44}$$

Least-squares FIR inverse filters are often used in many practical applications for deconvolution, including communications and seismic signal processing.

8.5.4 Design of IIR Filters in the Frequency Domain

The IIR filter design methods described in Sections 8.5.1 through 8.5.3 are carried out in the time domain. There are also direct design techniques for IIR filters that can be performed in the frequency domain. In this section we describe a filter parameter optimization technique carried out in the frequency domain that is representative of frequency-domain design methods.

The design is most easily carried out with the system function for the IIR filter expressed in the cascade form as

$$H(z) = G \prod_{k=1}^{K} \frac{1 + \beta_{k1} z^{-1} + \beta_{k2} z^{-2}}{1 + \alpha_{k1} z^{-1} + \alpha_{k2} z^{-2}} \tag{8.5.45}$$

where the filter gain G and the filter coefficients $\{\alpha_{k1}\}$, $\{\alpha_{k2}\}$, $\{\beta_{k1}\}$, $\{\beta_{k2}\}$ are to be determined. The frequency response of the filter can be expressed as

$$H(\omega) = GA(\omega)e^{j\Theta(\omega)} \tag{8.5.46}$$

where

$$A(\omega) = \prod_{k=1}^{K} \left| \frac{1 + \beta_{k1} z^{-1} + \beta_{k2} z^{-2}}{1 + \alpha_{k1} z^{-1} + \alpha_{k2} z^{-2}} \right|_{z=e^{j\omega}} \tag{8.5.47}$$

and $\Theta(\omega)$ is the phase response.

Instead of dealing with the phase of the filter, it is more convenient to deal with the envelope delay as a function of frequency, which is

$$\tau_g(\omega) = -\frac{d\Theta(\omega)}{d\omega} \tag{8.5.48}$$

or, equivalently,

$$\begin{aligned} \tau_g(\omega) &= \tau_g(z)|_{z=e^{j\omega}} \\ &= -\left[\frac{d\Theta(z)}{dz}\right]_{z=e^{j\omega}} \frac{dz}{d\omega} \end{aligned} \tag{8.5.49}$$

It can be shown that $\tau_g(z)$ can be expressed as

$$\tau_g(z) = \text{Re}\left\{ \sum_{k=1}^{K} \left[\frac{\beta_{k1} z + 2\beta_{k2}}{z^2 + \beta_{k1} z + \beta_{k2}} - \frac{\alpha_{k1} z + 2\alpha_{k2}}{z^2 + \alpha_{k1} z + \alpha_{k2}} \right] \right\} \tag{8.5.50}$$

where $Re(u)$ denotes the real part of the complex-valued quantity u.

Now suppose that the desired magnitude and delay characteristics $A(\omega)$ and $\tau_g(\omega)$ are specified at arbitrarily chosen discrete frequencies $\omega_1, \omega_2, \ldots, \omega_L$ in the range $0 \leq |\omega| \leq \pi$. Then the error in magnitude at the frequency ω_k is $GA(\omega_k) - A_d(\omega_k)$ where $A_d(\omega_k)$ is the desired magnitude response at ω_k. Similarly, the error in delay at ω_k can be defined as $\tau_g(\omega_k) - \tau_d(\omega_k)$, where $\tau_d(\omega_k)$ is the desired delay response. However, the choice of $\tau_d(\omega_k)$ is complicated by the difficulty in assigning a nominal delay to the filter. Hence, we are led to define the error in delay as $\tau_g(\omega_k) - \tau_g(\omega_0) - \tau_d(\omega_k)$, where $\tau_g(\omega_0)$ is the filter delay at some nominal center frequency in the passband of the filter and $\tau_d(\omega_k)$ is the desired delay response of the filter relative to $\tau_g(\omega_0)$. By defining the error in delay in this manner, we are willing to accept a filter having whatever nominal delay $\tau_g(\omega_0)$ results from the optimization procedure.

As a performance index for determining the filter parameters, one can choose any arbitrary function of the errors in magnitude and delay. To be specific, let us select the total weighted least-squares error over all frequencies $\omega_1, \omega_2, \ldots, \omega_L$, that is,

$$\begin{aligned}\mathcal{E}(\mathbf{p}, G) &= (1-\lambda)\sum_{n=1}^{L} w_n[GA(\omega_n) - A_d(\omega_n)]^2 \\ &\quad + \lambda \sum_{n=1}^{L} v_n[\tau_g(\omega_n) - \tau_g(\omega_0) - \tau_d(\omega_n)]^2 \end{aligned} \tag{8.5.51}$$

where $\mathbf{p}$ denotes the $4K$-dimensional vector of filter coefficients $\{\alpha_{k1}\}$, $\{\alpha_{k2}\}$, $\{\beta_{k1}\}$, and $\{\beta_{k2}\}$, and λ, $\{w_n\}$, and $\{v_n\}$ are weighting factors selected by the designer. Thus the emphasis on the errors affecting the design may be placed entirely on the magnitude ($\lambda = 0$), or on the delay ($\lambda = 1$) or, perhaps, equally weighted between magnitude and delay ($\lambda = 1/2$). Similarly, the weighting factors in frequency $\{w_n\}$ and $\{v_n\}$ determine the relative emphasis on the errors as a function of frequency.

The squared-error function E(**p**, G) is a nonlinear function of $(4K + 1)$ parameters. The gain G that minimizes $\mathcal{E}$ is easily determined and given by the relation

$$\hat{G} = \frac{\sum_{n=1}^{L} w_n A(\omega_n) A_d(\omega_n)}{\sum_{n=1}^{L} w_n A^2(\omega_n)} \tag{8.5.52}$$

The optimum gain G can be substituted in (8.5.51) to yield

$$\begin{aligned}\mathcal{E}(\mathbf{p}, \hat{G}) &= (1-\lambda)\sum_{n=1}^{L} w_n[\hat{G}A(\omega_n) - A_d(\omega_n)]^2 \\ &\quad + \lambda \sum_{n=1}^{L} v_n[\tau_g(\omega_n) - \tau_g(\omega_0) - \tau_d(\omega_n)]^2 \end{aligned} \tag{8.5.53}$$

Due to the nonlinear nature of $\mathcal{E}(\mathbf{p}, \hat{G})$, its minimization over the remaining $4K$ parameters is performed by an iterative numerical optimization method such as the Fletcher and Powell method (1963). One begins the iterative process by assuming an initial set of parameter values, say $\mathbf{p}^{(0)}$. With the initial values substituted in (8.5.51), we obtain the least-squares error $\mathcal{E}(\mathbf{p}^{(0)}, \hat{G})$. If we also evaluate the partial derivatives $\partial\mathcal{E}/\partial\alpha_{k1}$, $\partial\mathcal{E}/\partial\alpha_{k2}$, $\partial\mathcal{E}/\partial\beta_{k1}$, and $\partial\mathcal{E}/\partial\beta_{k2}$ at the initial value $\mathbf{p}^{(0)}$, we can use this first derivative information to change the initial values of the parameters in a direction that leads toward the minimum of the function $\mathcal{E}(\mathbf{p}, \hat{G})$ and thus to a new set of parameters $\mathbf{p}^{(1)}$.

Repetition of the above steps results in an iterative algorithm which is described mathematically by the recursive equation

$$\mathbf{p}^{(m+1)} = \mathbf{p}^{(m)} - \Delta^{(m)}\mathbf{Q}^{(m)}\mathbf{g}^{(m)} \qquad m = 0, 1, 2, \ldots$$

where $\Delta^{(m)}$ is a scalar representing the step size of the iteration, $\mathbf{Q}^{(m)}$ is a $(4K \times 4K)$ matrix, which is an estimate of the Hessian, and $\mathbf{g}^{(m)}$ is a $(4K \times 1)$ vector consisting of the four K-dimensional vectors of gradient components of $\mathcal{E}$ (i.e., $\partial\mathcal{E}/\partial\alpha_{k1}$, $\partial\mathcal{E}/\partial\alpha_{k2}$, $\partial\mathcal{E}/\partial\beta_{k1}$, $\partial\mathcal{E}/\partial\beta_{k2}$), evaluated at $\alpha_{k1} = \alpha_{k1}^{(m)}$, $\alpha_{k2} = \alpha_{k2}^{(m)}$, $\beta_{k1} = \beta_{k1}^{(m)}$, and $\beta_{k2} = \beta_{k2}^{(m)}$. This iterative process is terminated when the gradient components are nearly zero and the value of the function $\mathcal{E}(\mathbf{p}, \hat{G})$ does not change appreciably from one iteration to another.

The stability constraint is easily incorporated into the computer program through the parameter vector $\mathbf{p}$. When $|\alpha_{k2}| > 1$ for any $k = 1, \ldots, K$, the parameter α_{k2} is forced back inside the unit circle and the iterative process continued. A similar process can be used to force zeros inside the unit circle if a minimum-phase filter is desired.

The major difficulty with any iterative procedure that searches for the parameter values that minimize a nonlinear function is that the process may converge to a local minimum instead of a global minimum. Our only recourse around this problem is to start the iterative process with different values for the parameters and observe the end result.

Example 8.5.8

Let us design a lowpass filter using the Fletcher–Powell optimization procedure just described. The filter is to have a bandwidth of 0.3π and a rejection band commencing at 0.45π. The delay distortion can be ignored by selecting the weighting factor $\lambda = 0$.

Solution We have selected a two-stage ($K = 2$) or four-pole and four-zero filter which we believe is adequate to meet the transition band and rejection requirements. The magnitude response is specified at 19 equally spaced frequencies, which is considered a sufficiently dense set of points to realize a good design. Finally, a set of uniform weights is selected.

This filter has the response shown in Fig. 8.57. It has a remarkable resemblance to the response of the elliptic lowpass filter shown in Fig. 8.58, which was designed to have the same passband ripple and transition region as the computer-generated filter. A small but noticeable difference between the elliptic filter and the computer-generated filter is the somewhat flatter delay response of the latter relative to the former.

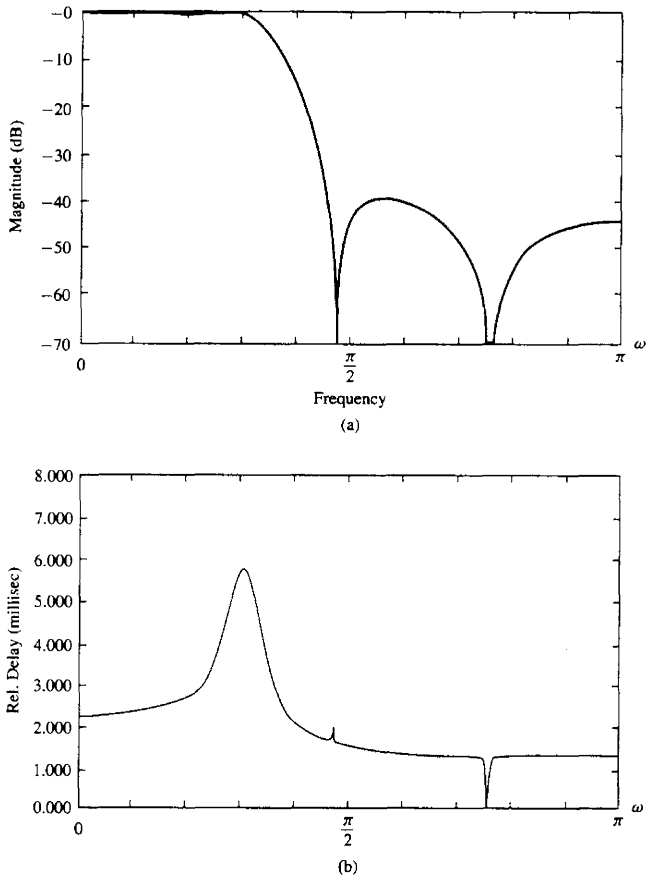


Figure 8.57 Filter designed by Fletcher–Powell optimization method (Example 8.5.8).

Example 8.5.9

Design an IIR filter with magnitude characteristics

$$A_d(\omega) = \begin{cases} \sin\omega, & 0 \le |\omega| \le \dfrac{\pi}{2} \\ 0, & \dfrac{\pi}{2} < |\omega| < \pi \end{cases}$$

and a constant envelope delay in the passband.

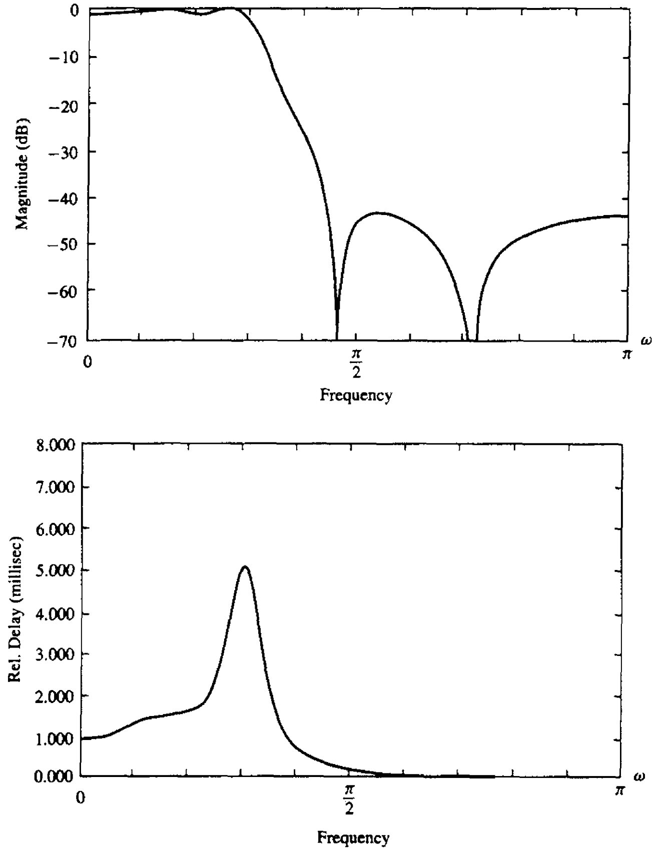


Figure 8.58 Amplitude and delay response for elliptic filter.

Solution The desired filter is called a modified duobinary filter and finds application in high-speed digital communications modems. The frequency response was specified at the frequencies illustrated in Fig. 8.59. The envelope delay was left unspecified in the stopband and selected to be flat in the passband. Equal weighting coefficients $\{w_n\}$ and $\{v_n\}$ were selected. A weighting factor of $\lambda = 1/2$ was selected.

A two-stage (four-pole, four-zero) filter is designed to meet the foregoing specifications. The result of the design is illustrated in Fig. 8.60. We note that the magnitude characteristic is reasonably well matched to $\sin \omega$ in the passband, but the stopband attenuation peaks at about −25 dB, which is rather large. The envelope delay characteristic is relatively flat in the passband.

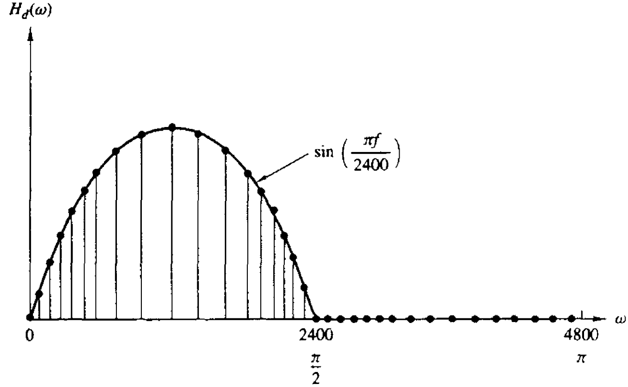


Figure 8.59 Frequency response of an ideal modified duobinary filter.

A four-stage (eight-pole, eight-zero) filter having the same frequency response specifications was also designed. This design produced better results, especially in the stopband where the attenuation peaked at −36 dB. The envelope delay was also considerably flatter.

8.6 SUMMARY AND REFERENCES

We have described in some detail the most important techniques for designing FIR and IIR digital filters based on either frequency-domain specifications expressed in terms of a desired frequency response $H_d(\omega)$, or in terms of the desired impulse response $h_d(n)$.

As a general rule, FIR filters are used in applications where there is a need for a linear-phase filter. This requirement occurs in many applications, especially in telecommunications, where there is a requirement to separate (demultiplex) signals such as data that have been frequency-division multiplexed, without distorting these signals in the process of demultiplexing. Of the several methods described for designing FIR filters, the frequency sampling design method and the optimum Chebyshev approximation method yield the best designs.

IIR filters are generally used in applications where some phase distortion is tolerable. Of the class of IIR filters, elliptic filters are the most efficient to implement in the sense that for a given set of specifications, an elliptic filter has a lower order or fewer coefficients than any other IIR filter type. When compared with FIR filters, elliptic filters are also considerably more efficient. In view of this, one might consider the use of an elliptic filter to obtain the desired frequency selectivity, followed then by an all-pass phase equalizer that compensates for the phase distortion in the elliptic filter. However, attempts to accomplish this have resulted in filters with a number of coefficients in the cascade combination that

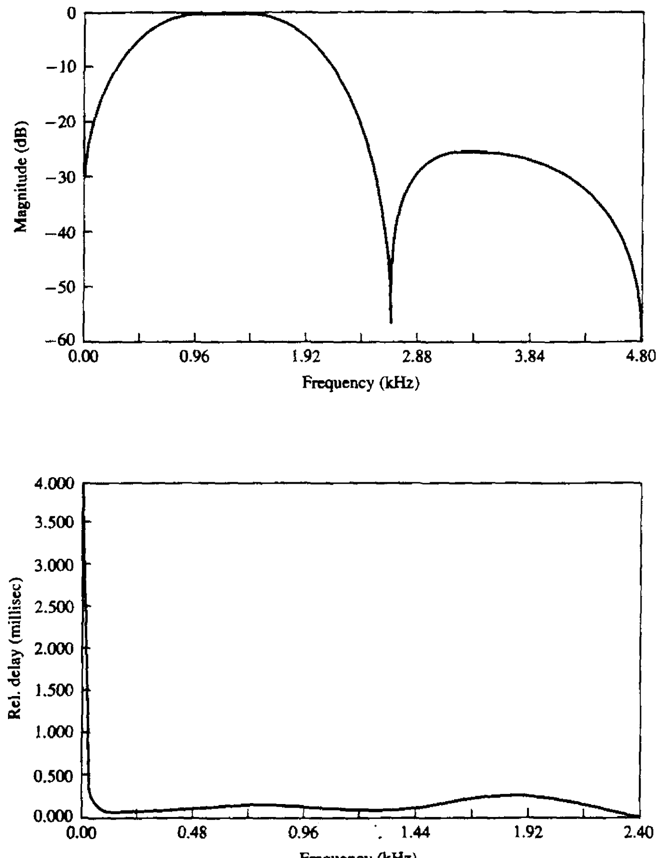


Figure 8.60 Frequency response of filter in Example 8.5.9. Designed by the Fletcher–Powell optimization method.

equaled or exceeded the number of coefficients in an equivalent linear-phase FIR filter. Consequently, no reduction in complexity is achievable in using phase-equalized elliptic filters.

In addition to the filter design methods based on the transformation of analog filters into the digital domain, we also presented several methods in which the design is done directly in the discrete-time domain. The least-squares method is particularly appropriate for designing IIR filters. The least-squares method is also used for the design of FIR Wiener filters.

Such a rich literature now exists on the design of digital filters that it is not possible to cite all the important references. We shall cite only a few. Some of the early work on digital filter design was done by Kaiser (1963, 1966), Steiglitz (1965), Golden and Kaiser (1964), Rader and Gold (1967a), Shanks (1967), Helms (1968), Gibbs (1969, 1970), and Gold and Rader (1969).

The design of analog filters is treated in the classic books by Storer (1957), Guillemin (1957), Weinberg (1962), and Daniels (1974).

The frequency sampling method for filter design was first proposed by Gold and Jordan (1968, 1969), and optimized by Rabiner et al. (1970). Additional results were published by Herrmann (1970), Herrmann and Schuessler (1970a), and Hofstetter et al. (1971). The Chebyshev (minimax) approximation method for designing linear-phase FIR filters was proposed by Parks and McClellan (1972a,b) and discussed further by Rabiner et al. (1975). The design of elliptic digital filters is treated in the book by Gold and Rader (1969) and in the paper by Gray and Markel (1976). The latter includes a computer program for designing digital elliptic filters.

The use of frequency transformations in the digital domain was proposed by Constantinides (1967, 1968, 1970). These transformations are appropriate only for IIR filters. The reader should note that when these transformations are applied to a lowpass FIR filter, the resulting filter is IIR.

Direct design techniques for digital filters have been considered in a number of papers, including Shanks (1967), Burrus and Parks (1970), Steiglitz (1970), Deczky (1972), Brophy and Salazar (1973), and Bandler and Bardakjian (1973).

PROBLEMS

8.1 Design an FIR linear phase, digital filter approximating the ideal frequency response

$$H_d(\omega) = \begin{cases} 1, & \text{for } |\omega| \le \dfrac{\pi}{6} \\ 0, & \text{for } \dfrac{\pi}{6} < |\omega| \le \pi \end{cases}$$

(a) Determine the coefficients of a 25-tap filter based on the window method with a rectangular window.
(b) Determine and plot the magnitude and phase response of the filter.
(c) Repeat parts (a) and (b) using the Hamming window.
(d) Repeat parts (a) and (b) using a Bartlett window.

8.2 Repeat Problem 8.1 for a bandstop filter having the ideal response

$$H_d(\omega) = \begin{cases} 1, & \text{for } |\omega| \le \dfrac{\pi}{6} \\ 0, & \text{for } \dfrac{\pi}{6} < |\omega| < \dfrac{\pi}{3} \\ 1, & \text{for } \dfrac{\pi}{3} \le |\omega| \le \pi \end{cases}$$

8.3 Redesign the filter of Problem 8.1 using the Hanning and Blackman windows.

8.4 Redesign the filter of Problem 8.2 using the Hanning and Blackman windows.

8.5 Determine the unit sample response $\{h(n)\}$ of a linear-phase FIR filter of length $M = 4$ for which the frequency response at $\omega = 0$ and $\omega = \pi/2$ is specified as

$$H_r(0) = 1 \qquad H_r\left(\frac{\pi}{2}\right) = \frac{1}{2}$$

8.6 Determine the coefficients $\{h(n)\}$ of a linear-phase FIR filter of length $M = 15$ which has a symmetric unit sample response and a frequency response that satisfies the condition

$$H_r\left(\frac{2\pi k}{15}\right) = \begin{cases} 1, & k = 0, 1, 2, 3 \\ 0, & k = 4, 5, 6, 7 \end{cases}$$

8.7 Repeat the filter design problem in Problem 8.6 with the frequency response specifications

$$H_r\left(\frac{2\pi k}{15}\right) = \begin{cases} 1 & k = 0, 1, 2, 3 \\ 0.4 & k = 4 \\ 0 & k = 5, 6, 7 \end{cases}$$

8.8 The ideal analog differentiator is described by

$$y_a(t) = \frac{dx_a(t)}{dt}$$

where $x_a(t)$ is the input and $y_a(t)$ the output signal.

(a) Determine its frequency response by exciting the system with the input $x_a(t) = e^{j2\pi Ft}$.

(b) Sketch the magnitude and phase response of an ideal analog differentiator band-limited to B hertz.

(c) The ideal digital differentiator is defined as

$$H(\omega) = j\omega \qquad |\omega| \leq \pi$$

Justify this definition by comparing the frequency response $|H(\omega)|$, $\measuredangle H(\omega)$ with that in part (b).

(d) By computing the frequency response $H(\omega)$, show that the discrete-time system

$$y(n) = x(n) - x(n-1)$$

is a good approximation of a differentiator at low frequencies.

(e) Compute the response of the system to the input

$$x(n) = A\cos(\omega_0 n + \theta)$$

8.9 Use the window method with a Hamming window to design a 21-tap differentiator as shown in Fig. P8.9. Compute and plot the magnitude and phase response of the resulting filter.

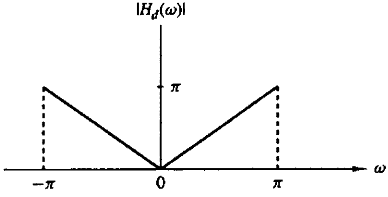


Figure P8.9

8.10 Use the matched-z transformation to convert the analog filter with system function

$$H(s) = \frac{s+0.1}{(s+0.1)^2+9}$$

into a digital IIR filter. Select $T = 0.1$ and compare the location of the zeros in $H(z)$ with the locations of the zeros obtained by applying the impulse invariance method in the conversion of $H(s)$.

8.11 Convert the analog bandpass filter designed in Example 8.4.1 into a digital filter by means of the bilinear transformation. Thereby derive the digital filter characteristic obtained in Example 8.4.2 by the alternative approach and verify that the bilinear transformation applied to the analog filter results in the same digital bandpass filter.

8.12 An ideal analog integrator is described by the system function $H_a(s) = 1/s$. A digital integrator with system function $H(z)$ can obtained by use of the bilinear transformation. That is,

$$H(z) = \frac{T}{2}\frac{1+z^{-1}}{1-z^{-1}} \equiv H_a(s)|_{s=(2/T)(1-z^{-1})/(1+z^{-1})}$$

(a) Write the difference equation for the digital integrator relating the input $x(n)$ to the output $y(n)$.

(b) Roughly sketch the magnitude $|H_a(j\Omega)|$ and phase $\Theta(\Omega)$ of the analog integrator.

(c) It is easily verified that the frequency response of the digital integrator is

$$H(\omega) = -j\frac{T}{2}\frac{\cos(\omega/2)}{\sin(\omega/2)} = -j\frac{T}{2}\cot\frac{\omega}{2}$$

Roughly sketch $|H(\omega)|$ and $\theta(\omega)$.

(d) Compare the magnitude and phase characteristics obtained in parts (b) and (c). How well does the digital integrator match the magnitude and phase characteristics of the analog integrator?

(e) The digital integrator has a pole at $z = 1$. If you implement this filter on a digital computer, what restrictions might you place on the input signal sequence $x(n)$ to avoid computational difficulties?

8.13 A z-plane pole–zero plot for a certain digital filter is shown in Fig. P8.13. The filter has unity gain at dc.

(a) Determine the system function in the form

$$H(z) = A\left[\frac{(1+a_1z^{-1})(1+b_1z^{-1}+b_2z^{-2})}{(1+c_1z^{-1})(1+d_1z^{-1}+d_2z^{-2})}\right]$$

giving numerical values for the parameters A, a_1, b_1, b_2, c_1, d_1, and d_2.

(b) Draw block diagrams showing numerical values for path gains in the following forms:

(1) Direct form II (canonic form)

(2) Cascade form (make each section canonic, with real coefficients)

8.14 Consider the pole–zero plot shown in Fig. P8.14.

(a) Does it represent an FIR filter?

(b) Is it a linear-phase system?

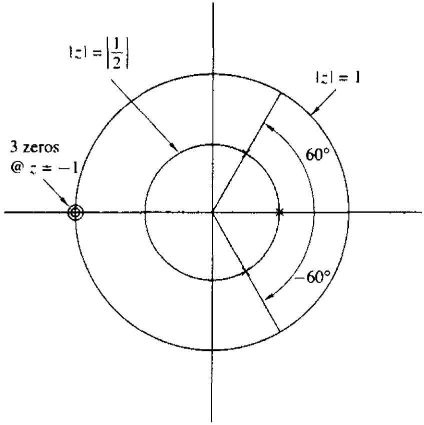


Figure P8.13

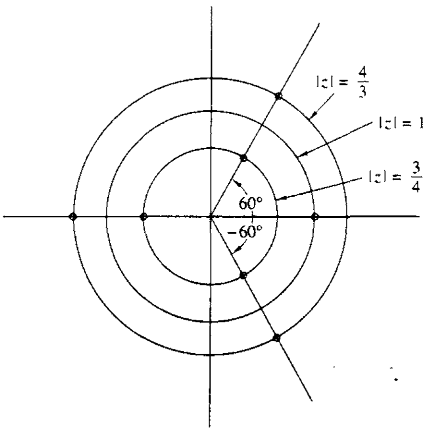


Figure P8.14

(c) Give a direct form realization that exploits all symmetries to minimize the number of multiplications. Show all path gains.

8.15* A digital low-pass filter is required to meet the following specifications:

Passband ripple: ≤ 1 dB
Passband edge: 4 kHz
Stopband attenuation: ≥ 40 dB
Stopband edge: 6 kHz
Sample rate: 24 kHz

The filter is to be designed by performing a bilinear transformation on an analog system function. Determine what order Butterworth, Chebyshev, and elliptic analog designs must be used to meet the specifications in the digital implementation.

8.16* An IIR digital low-pass filter is required to meet the following specifications:

Passband ripple (or peak-to-peak ripple): ≤ 0.5 dB
Passband edge: 1.2 kHz
Stopband attenuation: ≥ 40 dB
Stopband edge: 2.0 kHz
Sample rate: 8.0 kHz

Use the design formulas in the book to determine the required filter order for

(a) A digital Butterworth filter
(b) A digital Chebyshev filter
(c) A digital elliptic filter

8.17* Determine the system function $H(z)$ of the lowest-order Chebyshev digital filter that meets the following specifications:

(a) 1-dB ripple in the passband $0 \leq |\omega| \leq 0.3\pi$.
(b) At least 60 dB attentuation in the stopband $0.35\pi \leq |\omega| \leq \pi$. Use the bilinear transformation.

8.18* Determine the system function $H(z)$ of the lowest-order Chebyshev digital filter that meets the following specifications:

(a) $\frac{1}{2}$-dB ripple in the passband $0 \leq |\omega| \leq 0.24\pi$.
(b) At least 50-dB attenuation in the stopband $0.35\pi \leq |\omega| \leq \pi$. Use the bilinear transformation.

8.19* An analog signal $x(t)$ consists of the sum of two components $x_1(t)$ and $x_2(t)$. The spectral characteristics of $x(t)$ are shown in the sketch in Fig. P8.19. The signal $x(t)$ is bandlimited to 40 kHz and it is sampled at a rate of 100 kHz to yield the sequence $x(n)$.

It is desired to suppress the signal $x_2(t)$ by passing the sequence $x(n)$ through a digital lowpass filter. The allowable amplitude distortion on $|X_1(f)|$ is $\pm 2\%$ $(\delta_1 = 0.02)$ over the range $0 \leq |F| \leq 15$ kHz. Above 20 kHz, the filter must have an attenuation of at least 40 dB $(\delta_2 = 0.01)$.

(a) Use the Remez exchange algorithm to design the *minimum*-order linear-phase FIR filter that meets the specifications above. From the plot of the magnitude characteristic of the filter frequency response, give the actual specifications achieved by the filter.
(b) Compare the order M obtained in part (a) with the approximate formulas given in equations (8.2.94) and (8.2.95).
(c) For the order M obtained in part (a), design an FIR digital lowpass filter using the window technique and the Hamming window. Compare the frequency response characteristics of this design with those obtained in part (a).

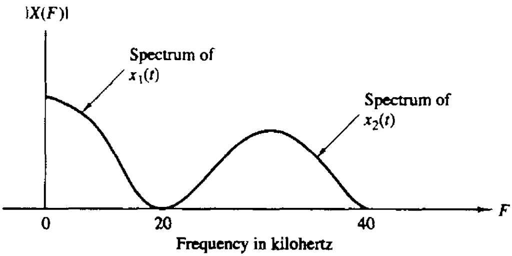


Figure P8.14

(d) Design the *minimum*-order elliptic filter that meets the given amplitude specifications. Compare the frequency response of the elliptic filter with that of the FIR filter in part (a).

(e) Compare the complexity of implementing the FIR filter in part (a) versus the elliptic filter obtained in part (d). Assume that the FIR filter is implemented in the direct form and the elliptic filter is implemented as a cascade of two-pole filters. Use storage requirements and the number of multiplications per output point in the comparison of complexity.

8.20 The impulse response of an analog filter is shown in Fig. P8.20.

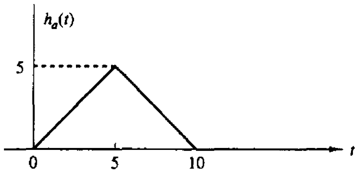


Figure P8.20

(a) Let $h(n) = h_a(nT)$, where $T = 1$, be the impulse response of a discrete-time filter. Determine the system function $H(z)$ and the frequency response $H(\omega)$ for this FIR filter.

(b) Sketch (roughly) $|H(\omega)|$ and compare this frequency response characteristic with $|H_a(j\Omega)|$.

(c) The FIR filter with unit sample response $h(n)$ given above is to be approximated by a second-order IIR filter of the form

$$G(z) = \frac{b_0 z^{-1}}{1 - a_1 z^{-1} - a_2 z^{-2}}$$

Use the least-squares inverse design procedure to determine the values of the coefficients b_0, a_1, and a_2.

8.21 In this problem you will be comparing some of the characteristics of analog and digital implementations of the single-pole low-pass analog system

$$H_a(s) = \frac{1}{s+\alpha} \Leftrightarrow h_a(t) = e^{-\alpha t}$$

(a) What is the gain at dc? At what radian frequency is the analog frequency response 3 dB down from its dc value? At what frequency is the analog frequency reponse zero? At what time has the analog impulse response decayed to $1/e$ of its initial value?

(b) Give the digital system function $H(z)$ for the impulse-invariant design for this filter. What is the gain at dc? Give an expression for the 3-dB radian frequency. At what (real-valued) frequency is the response zero? How many samples are there in the unit sample time-domain response before it has decayed to $1/e$ of its initial value?

(c) "Prewarp" the parameter α and perform the bilinear transformation to obtain the digital system function $H(z)$ from the analog design. What is the gain at dc? At what (real-valued) frequency is the response zero? Give an expression for the 3-dB radian frequency. How many samples in the unit sample time-domain response before it has decayed to $1/e$ of its initial value?

8.22 We wish to design a FIR bandpass filter having a duration $M = 201$. $H_d(\omega)$ represents the ideal characteristic of the noncausal bandpass filter as shown in Fig. P8.22.

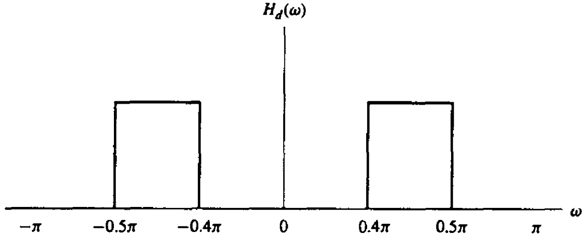


Figure P8.22

(a) Determine the unit sample (impulse) response $h_d(n)$ corresponding to $H_d(\omega)$.

(b) Explain how you would use the Hamming window

$$w(n) = 0.54 + 0.46\cos\left(\frac{2\pi}{M-1}\right) \qquad -\frac{M-1}{2} \le n \le \frac{M-1}{2}$$

to design a FIR bandpass filter having an impulse response $h(n)$ for $0 \le n \le 200$.

(c) Suppose that you were to design the FIR filter with $M = 201$ by using the frequency sampling technique in which the DFT coefficients $H(k)$ are specified instead of $h(n)$. Give the values of $H(k)$ for $0 \le k \le 200$ corresponding to $H_d(e^{j\omega})$ and indicate how the frequency response of the actual filter will differ from the ideal. Would the actual filter represent a good design? Explain your answer.

8.23 We wish to design a digital bandpass filter from a second-order analog lowpass Butterworth filter prototype using the bilinear transformation. The specifications on the digital filter are shown in Fig. P8.23(a). The cutoff frequencies (measured at the half power points) for the digital filter should lie at $\omega = 5\pi/12$ and $\omega = 7\pi/12$.

The analog protoype is given by

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

with the half-power point at $\Omega = 1$.

(a) Determine the system function for the digital bandpass filter.

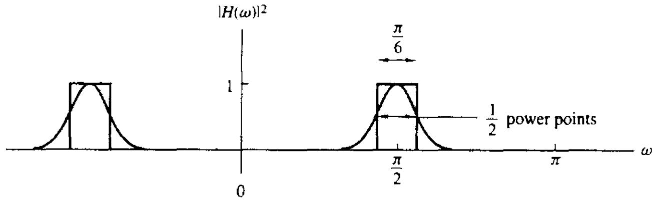
(b) Using the same specs on the digital filter as in part (a), determine which of the analog bandpass prototype filters shown in Fig. P.8.23(b) could be transformed directly using the bilinear transformation to give the proper digital filter. Only the plot of the magnitude squared of the frequency is given.

8.24 Figure P8.24 shows a digital filter designed using the frequency sampling method.

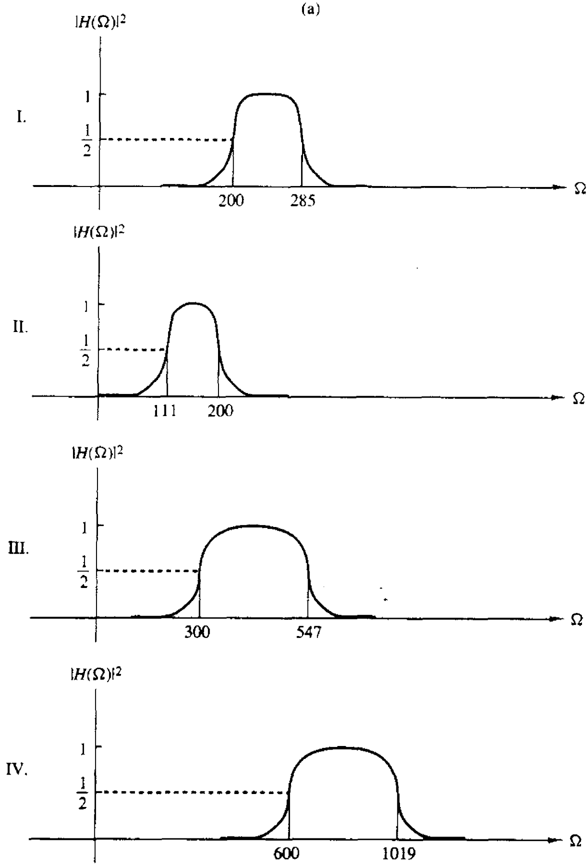
(a) Sketch a z-plane pole–zero plot for this filter.

(b) Is the filter lowpass, highpass, or bandpass?

(c) Determine the magnitude response $|H(\omega)$ at the frequencies $\omega_k = \pi k/6$ for $k = 0$, 1, 2, 3, 4, 5, 6.



(a)



(b)

Figure P8.23

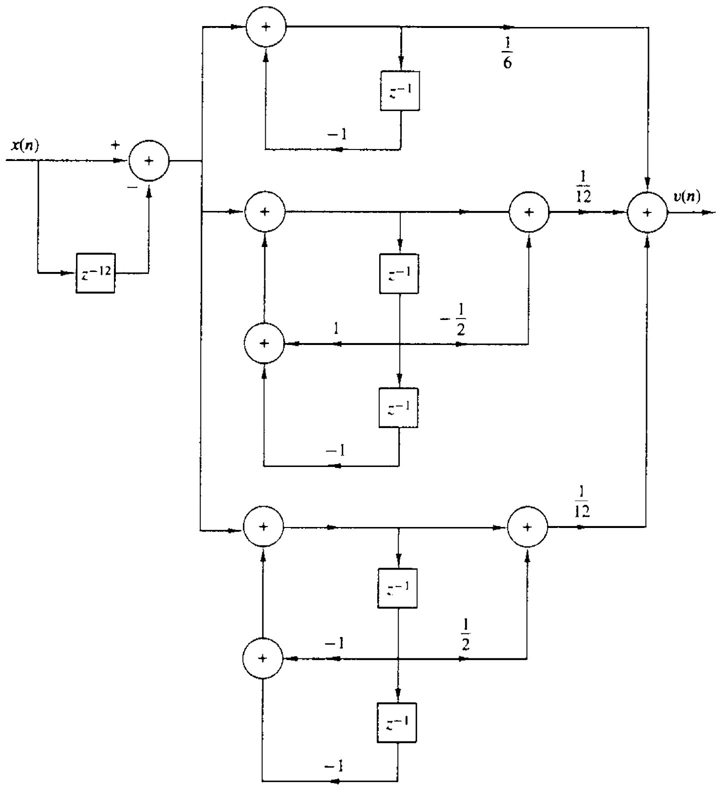


Figure P8.24

(d) Use the results of part (c) to sketch the magnitude response for $0 \le \omega \le \pi$ and confirm your answer to part (b).

8.25 An analog signal of the form $x_a(t) = a(t)\cos 2000\pi t$ is bandlimited to the range $900 \le F \le 1100$ Hz. It is used as an input to the system shown in Fig. P8.25.

(a) Determine and sketch the spectra for the signals $x(n)$ and $w(n)$.

(b) Use a Hamming window of length $M = 31$ to design a lowpass linear phase FIR filter $H(\omega)$ that passes $\{a(n)\}$.

(c) Determine the sampling rate of the A/D converter that would allow us to eliminate the frequency conversion in Fig. P8.25.

8.26 *System identification* Consider an unknown LTI system and an FIR system model as shown in Fig. P8.26. Both systems are excited by the same input sequence $\{x(n)\}$. The problem is to determine the coefficients $\{h(n), 0 \le n \le M - 1\}$ of the FIR model

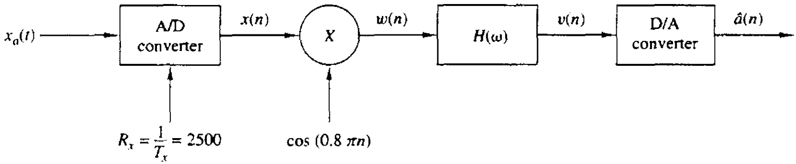


Figure P8.25

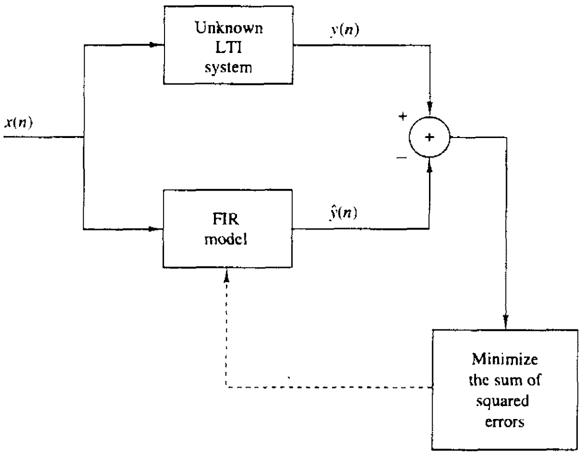


Figure P8.26

of the system to minimize the average squared error between the outputs of the two systems.

(a) Use the least-squares criterion to determine the equations for the optimum FIR filter coefficients.

(b) Repeat part (a) if the output of the unknown system is corrupted by an additive white noise $\{w(n)\}$ sequence with variance σ_w^2.

8.27 Determine the least-squares FIR inverse of length 3 to the system with impulse response

$$h(n) = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \\ 0, & \text{otherwise} \end{cases}$$

Also, determine the minimum squared error $\mathcal{E}_{\min}$.

8.28 Determine the least-squares FIR inverse filter of length 3 for the system with impulse response $h(n)$ given in Example 8.5.6, when $\alpha = \frac{1}{2}$ and the desired response is specified as $d(n) = \delta(n-2)$. Also compute the minimum least-squares error.

8.29* A linear time-invariant system has an input sequence $x(n)$ and an output sequence $y(n)$. The user has access only to the system output $y(n)$. In addition, the following information is available.

(a) The input signal is periodic with a given fundamental period N and has a flat spectral envelope, that is,

$$x(n) = \sum_{k=0}^{N-1} c_k^x e^{j(2\pi/N)kn} \qquad \text{all } n$$

where $c_k^x = 1$ for all k.

(b) The system $H(z)$ is all-pole, that is,

$$H(z) = \frac{1}{1 + \sum_{k=1}^{p} a_k z^{-k}}$$

but the order p and the coefficients $(a_k, 1 \le k \le p)$ are unknown. Is it possible to determine the order p and the numerical values of the coefficients $\{a_k, 1 \le k \le p\}$ by taking measurements on the output $y(n)$? If yes, explain how. Is this possible for every value of p?

(c) Repeat Problem 8.31 for a system with system function

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{p} a_k z^{-k}}$$

(d) *FIR system modeling* Consider an "unknown" FIR system with impulse response $h(n)$, $0 \le n \le 11$, given by

$$\begin{aligned}
h(0) = h(11) &= 0.309828 \times 10^{-1} \\
h(1) = h(10) &= 0.416901 \times 10^{-1} \\
h(2) = h(9) &= -0.577081 \times 10^{-1} \\
h(3) = h(8) &= -0.852502 \times 10^{-1} \\
h(4) = h(7) &= 0.147157 \times 10^{0} \\
h(5) = h(6) &= 0.449188 \times 10^{0}
\end{aligned}$$

A potential user has access to the input and output of the system but does not have any information about its impulse response other than that it is FIR. In an effort to determine the impulse response of the system, the user excites it with a zero mean, random sequence $x(n)$ uniformly distributed in the range $[-0.5, 0.5]$, and records the signal $x(n)$ and the corresponding output $y(n)$ for $0 \le n \le 199$.

(1) By using the available information that the unknown system is FIR, the user employs the method of least-squares to obtain an FIR model $h(n)$, $0 \le n \le M - 1$. Set up the system of linear equations, specifying the parameters $h(0), h(1), \ldots, h(M-1)$. Specify formulas we should use to determine the necessary autocorrelation and crosscorrelation values.

(2) Since the order of the system is unknown, the user decides to try models of different orders and check the corresponding total squared error. Clearly, this error will be zero (or very close to it if the order of the model becomes equal to the order of the system). Compute the FIR models $h_M(n)$, $0 \leq n \leq M-1$ for $M = 8, 9, 10, 11, 12, 13, 14$ as well as the corresponding total squared errors E_M, $M = 8, 9, \ldots, 14$. What do you observe?

(3) Determine and plot the frequency response of the system and the models for $M = 11, 12, 13$. Comment on the results.

(4) Suppose now that the output of the system is corrupted by additive noise, so instead of the signal $y(n)$, $0 \leq n \leq 199$, we have available the signal

$$v(n) = y(n) + 0.01w(n)$$

where $w(n)$ is a Gaussian random sequence with zero mean and variance $\sigma^2 = 1$.

Repeat part (b) by using $v(n)$ instead of $y(n)$ and comment on the results. The quality of the model can be also determined by the quantity

$$Q = \frac{\displaystyle\sum_{n=0}^{\infty}[h(n) - \hat{h}(n)]^2}{\displaystyle\sum_{n=0}^{\infty} h^2(n)}$$

9

Sampling and Reconstruction of Signals

In Chapters 1 and 4 we treated the sampling of continuous-time signals and demonstrated that if the signals are bandlimited, it is possible to reconstruct the original signal from the samples, provided that the sampling rate is at least twice the highest frequency contained in the signal. We also briefly described the subsequent operations of quantization and coding that are necessary to convert an analog signal to a digital signal appropriate for digital processing.

In this chapter we consider time-domain sampling, analog-to-digital (A/D) conversion (quantization and coding), and digital-to-analog (D/A) conversion (signal reconstruction) in greater depth. First, we consider the sampling of the special class of signals that are characterized as bandpass signals. Then we treat analog-to-digital converters and their characteristics. Of particular interest is the use of oversampling and sigma-delta modulation in the design of high precision A/D converters. The final topic of the chapter is digital-to-analog conversion or, simply, the reconstruction of the continuous-time signal from its sampled values.

9.1 SAMPLING OF BANDPASS SIGNALS

Our main focus in this section is the sampling of bandpass signals. We begin by describing the time and frequency domain representations of bandpass signals.

9.1.1 Representation of Bandpass Signals

Suppose that a real-valued signal $x(t)$ has a frequency content concentrated in a narrow band of frequencies in the vicinity of a frequency F_c, as shown in Fig. 9.1. Our objective is to develop a mathematical representation of such signals. First, we construct a signal that contains only the positive frequencies in $x(t)$. Such a signal can be expressed as

$$X_+(F) = 2V(F)X(F) \tag{9.1.1}$$

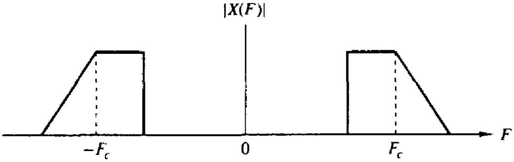


Figure 9.1 Spectrum of a bandpass signal.

where $X(F)$ is the Fourier transform of $x(t)$ and $V(F)$ is the unit step function. The equivalent time-domain expression for (9.1.1) is

$$\begin{aligned} x_+(t) &= \int_{-\infty}^{\infty} X_+(F)e^{j2\pi Ft}dF \\ &= F^{-1}[2V(F)] \star F^{-1}[X(F)] \end{aligned} \tag{9.1.2}$$

The signal $x_+(t)$ is called the *analytic signal* or the *pre-envelope* of $x(t)$. We note that $F^{-1}[X(F)] = x(t)$ and

$$F^{-1}[2V(f)] = \delta(t) + \frac{j}{\pi t} \tag{9.1.3}$$

Hence,

$$\begin{aligned} x_+(t) &= \left[\delta(t) + \frac{j}{\pi t}\right] \star x(t) \\ &= x(t) + j\frac{1}{\pi t} \star x(t) \end{aligned} \tag{9.1.4}$$

We define $\hat{x}(t)$ as

$$\begin{aligned} \hat{x}(t) &= \frac{1}{\pi t} \star x(t) \\ &= \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\tau)}{t-\tau}d\tau \end{aligned} \tag{9.1.5}$$

The signal $\hat{x}(t)$ can be viewed as the output of the filter with impulse response

$$h(t) = \frac{1}{\pi t}, \qquad -\infty < t < \infty \tag{9.1.6}$$

when excited by the input signal $x(t)$. Such a filter is called a *Hilbert transformer*. The frequency response of this filter is simply

$$\begin{aligned} H(F) &= \int_{-\infty}^{\infty} h(t)e^{-j2\pi Ft}dt \\ &= \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{1}{t}e^{-2\pi Ft}dt \\ &= \begin{cases} -j & (F > 0) \\ 0 & (F = 0) \\ j & (F < 0) \end{cases} \end{aligned} \tag{9.1.7}$$

We observe that $|H(F)| = 1$ and that the phase response $\Theta(F) = -\frac{1}{2}\pi$ for $F > 0$

and $\Theta(F) = \frac{1}{2}\pi$ for $F < 0$. Therefore, this filter is basically a 90° phase shifter for all frequencies in the input signal, and it is akin to the discrete-time Hilbert transform filter described in Section 8.2.6.

The analytic signal $x_+(t)$ is a bandpass signal. We can obtain an equivalent lowpass representation by performing a frequency translation of $X_+(F)$. Thus, we define $X_l(F)$ as

$$X_l(F) = X_+(F + F_c) \tag{9.1.8}$$

The equivalent time-domain relation is

$$\begin{aligned} x_l(t) &= x_+(t)e^{-j2\pi F_c t} \\ &= [x(t) + j\hat{x}(t)]e^{-j2\pi F_c t} \end{aligned} \tag{9.1.9}$$

or, equivalently,

$$x(t) + j\hat{x}(t) = x_l(t)e^{j2\pi F_c t} \tag{9.1.10}$$

In general, the signal $x_l(t)$ is complex-valued (see Problem 9.3), and can be expressed as

$$x_l(t) = u_c(t) + ju_s(t) \tag{9.1.11}$$

If we substitute for $x_l(t)$ in (9.1.10) and equate real and imaginary parts on each side, we obtain the relations

$$x(t) = u_c(t)\cos 2\pi F_c t - u_s(t)\sin 2\pi F_c t \tag{9.1.12}$$

$$\hat{x}(t) = u_c(t)\sin 2\pi F_c t + u_s(t)\cos 2\pi F_c t \tag{9.1.13}$$

The expression (9.1.12) is the desired form for the representation of a bandpass signal. The low-frequency signal components $u_c(t)$ and $u_s(t)$ can be viewed as amplitude modulations impressed on the carrier components $\cos 2\pi F_c t$ and $\sin 2\pi F_c t$, respectively. Since these carrier components are in phase quadrature, $u_c(t)$ and $u_s(t)$ are called the *quadrature components* of the bandpass signal $x(t)$.

Another representation of the signal in (9.1.12) is

$$\begin{aligned} x(t) &= \mathrm{Re}\{[u_c(t) + ju_s(t)]e^{j2\pi F_c t}\} \\ &= \mathrm{Re}[x_l(t)e^{j2\pi F_c t}] \end{aligned} \tag{9.1.14}$$

where Re denotes the real part of the complex-valued quantity in the brackets following. The lowpass signal $x_l(t)$ is usually called the *complex envelope* of the real signal $x(t)$, and is basically the *equivalent lowpass signal.*

Finally, a third possible representation of a bandpass signal is obtained by expressing $x_l(t)$ as

$$x_l(t) = a(t)e^{j\theta(t)} \tag{9.1.15}$$

where

$$a(t) = \sqrt{u_c^2(t) + u_s^2(t)} \tag{9.1.16}$$

$$\theta(t) = \tan^{-1}\frac{u_s(t)}{u_c(t)} \tag{9.1.17}$$

Then

$$\begin{aligned} x(t) &= \mathrm{Re}[x_l(t)e^{j2\pi F_c t}] \\ &= \mathrm{Re}[a(t)e^{j[2\pi F_c t+\theta(t)]}] \\ &= a(t)\cos[2\pi F_c t + \theta(t)] \end{aligned} \tag{9.1.18}$$

The signal $a(t)$ is called the *envelope* of $x(t)$, and $\theta(t)$ is called the *phase* of $x(t)$. Therefore, (9.1.12), (9.1.14), and (9.1.18) are equivalent representations of bandpass signals.

The Fourier transform of $x(t)$ is

$$\begin{aligned} X(F) &= \int_{-\infty}^{\infty} x(t)e^{-j2\pi Ft}\,dt \\ &= \int_{-\infty}^{\infty} \{\mathrm{Re}[x_l(t)e^{j2\pi F_c t}]\}e^{-j2\pi Ft}\,dt \end{aligned} \tag{9.1.19}$$

Use of the identity

$$\mathrm{Re}(\xi) = \tfrac{1}{2}(\xi + \xi^*) \tag{9.1.20}$$

in (9.1.19) yields the result

$$\begin{aligned} X(F) &= \frac{1}{2}\int_{-\infty}^{\infty} [x_l(t)e^{j2\pi F_c t} + x_l^*(t)e^{-j2\pi F_c t}]e^{-j2\pi Ft}\,dt \\ &= \tfrac{1}{2}[X_l(F - F_c) + X_l^*(-F - F_c)] \end{aligned} \tag{9.1.21}$$

where $X_l(F)$ is the Fourier transform of $x_l(t)$. This is the basic relationship between the spectrum of the real bandpass signal $x(t)$ and the spectrum of the equivalent lowpass signal $x_l(t)$.

It is apparent from (9.1.21) that the spectrum of the bandpass signal $x(t)$ can be obtained from the spectrum of the complex signal $x_l(t)$ by a frequency translation. To be more precise, suppose that the spectrum of the signal $x_l(t)$ is as shown in Fig. 9.2(a). Then the spectrum $X(F)$ for positive frequencies is simply $X_l(F)$ translated in frequency to the right by F_c and scaled in amplitude by $\frac{1}{2}$. The spectrum $X(F)$ for negative frequencies is obtained by first folding $X_l(F)$ about $F = 0$ to obtain $X_l(-F)$, conjugating $X_l(-F)$ to obtain $X_l^*(-F)$, translating $X_l^*(-F)$ in frequency to the left by F_c, and scaling the result by $\frac{1}{2}$. The folding and conjugation of $X_l(F)$ for the negative-frequency component of the spectrum result in a magnitude spectrum $|X(F)|$ that is even and a phase spectrum $\measuredangle X(F)$ that is odd as shown in Fig. 9.2(b). These symmetry properties must hold since the signal $x(t)$ is real valued. However, they do not apply to the spectrum of the equivalent complex signal $X_l(t)$.

The development above implies that *any bandpass signal $x(t)$ can be represented by an equivalent lowpass signal $x_l(t)$*. In general, the equivalent lowpass signal $x_l(t)$ is complex valued, whereas the bandpass signal $x(t)$ is real. The latter can be obtained from the former through the time-domain relation in (9.1.14) or through the frequency-domain relation in (9.1.21).

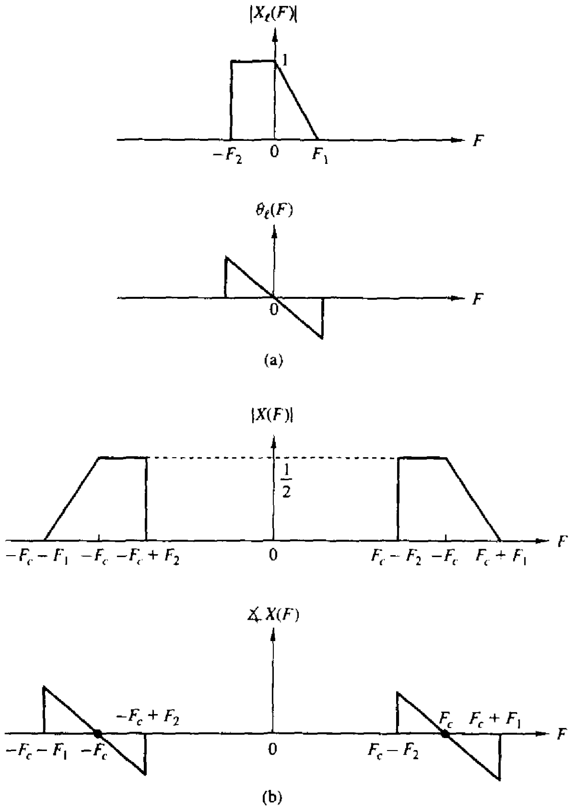


Figure 9.2 (a) Spectrum of the lowpass signal and (b) the corresponding spectrum for the bandpass signal.

9.1.2 Sampling of Bandpass Signals

We have already demonstrated that a continuous-time signal with highest frequency B can be uniquely represented by samples taken at the minimum rate (Nyquist rate) of $2B$ samples per second. However, if the signal is a bandpass signal with frequency components in the band $B_1 \leq F \leq B_2$, as shown in Fig. 9.3, a blind application of the sampling theorem would have us sampling the signal at a rate of $2B_2$ samples per second.

If that were the case and B_2 was an extremely high frequency, it would certainly be advantageous to perform a frequency shift of the bandpass signal by

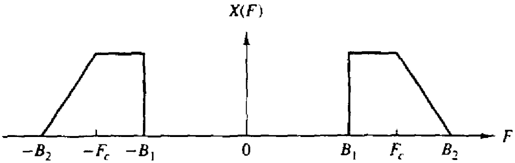


Figure 9.3 Bandpass signal with frequency components in the range $B_1 \le F \le B_2$.

an amount

$$F_c = \frac{B_1 + B_2}{2} \tag{9.1.22}$$

and sampling the equivalent lowpass signal. Such a frequency shift can be achieved by multiplying the bandpass signal as given in (9.1.12) by the quadrature carriers $\cos 2\pi F_c t$ and $\sin 2\pi F_c t$ and lowpass filtering the products to eliminate the signal components at $2F_c$. Clearly, the multiplication and the subsequent filtering are first performed in the analog domain and then the outputs of the filters are sampled. The resulting equivalent lowpass signal has a bandwidth $B/2$, where $B = B_2 - B_1$. Therefore, it can be represented uniquely by samples taken at the rate of B samples per second for each of the quadrature components. Thus the sampling can be performed on each of the lowpass filter outputs at the rate of B samples per second, as indicated in Fig. 9.4. Therefore, the resulting rate is $2B$ samples per second.

In view of the fact that frequency conversion to lowpass allows us to reduce the sampling rate to $2B$ samples per second, it should be possible to sample the bandpass signal at a comparable rate. In fact, it is.

Suppose that the upper frequency $F_c + B/2$ is a multiple of the bandwidth B (i.e., $F_c + B/2 = kB$), where k is a positive integer. If we sample $x(t)$ at the rate

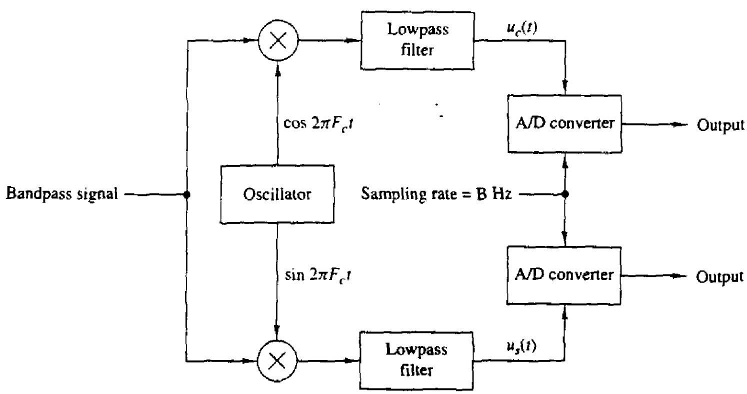


Figure 9.4 Sampling of a bandpass signal by first converting to an equivalent low-pass signal.

$2B = 1/T$ samples per second, we have

$$\begin{aligned} x(nT) &= u_c(nT)\cos 2\pi F_c nT - u_s(nT)\sin 2\pi F_c nT \\ &= u_c(nT)\cos\frac{\pi n(2k-1)}{2} - u_s(nT)\sin\frac{\pi n(2k-1)}{2} \end{aligned} \tag{9.1.23}$$

where the last step is obtained by substituting $F_c = kB - B/2$ and $T = 1/2B$.

For n even, say $n = 2m$, (9.1.23) reduces to

$$x(2mT) \equiv x(mT_1) = u_c(mT_1)\cos\pi m(2k-1) = (-1)^m u_c(mT_1) \tag{9.1.24}$$

where $T_1 = 2T = 1/B$. For n odd, say $n = 2m - 1$, (9.1.23) reduces to

$$x(2mT - T) \equiv x\left(mT_1 - \frac{T_1}{2}\right) = u_s\left(mT_1 - \frac{T_1}{2}\right)(-1)^{m+k+1} \tag{9.1.25}$$

Therefore, the even-numbered samples of $x(t)$, which occur at the rate of B samples per second, produce samples of the lowpass signal component $u_c(t)$. The odd-numbered samples of $x(t)$, which also occur at the rate of B samples per second, produce samples of the lowpass signal component $u_s(t)$.

Now, the samples $\{u_c(mT_1)\}$ and the samples $\{u_s(mT_1 - T_1/2)\}$ can be used to reconstruct the equivalent lowpass signals. Thus, according to the sampling theorem for lowpass signals with $T_1 = 1/B$,

$$u_c(t) = \sum_{m=-\infty}^{\infty} u_c(mT_1)\frac{\sin(\pi/T_1)(t - mT_1)}{(\pi/T_1)(t - mT_1)} \tag{9.1.26}$$

$$u_s(t) = \sum_{m=-\infty}^{\infty} u_s\left(mT_1 - \frac{T_1}{2}\right)\frac{\sin(\pi/T_1)(t - mT_1 + T_1/2)}{(\pi T_1)(t - mT_1 + T_1/2)} \tag{9.1.27}$$

Furthermore, the relations in (9.1.24) and (9.1.25) allow us to express $u_c(t)$ and $u_s(t)$ directly in terms of samples of $x(t)$. Now, since $x(t)$ is expressed as

$$x(t) = u_c(t)\cos 2\pi F_c t - u_s(t)\sin 2\pi F_c t \tag{9.1.28}$$

substitution from (9.1.27), (9.1.26), (9.1.25), and (9.1.24) into (9.1.28) yields

$$\begin{aligned} x(t) = \sum_{m=-\infty}^{\infty} \Bigg\{ &(-1)^m x(2mT)\frac{\sin(\pi/2T)(t - 2mT)}{(\pi/2T)(t - 2mT)}\cos 2\pi F_c t \\ &+ (-1)^{m+k} x((2m-1)T)\frac{\sin(\pi/2T)(t - 2mT + T)}{(\pi/2T)(t - 2mT + T)}\sin 2\pi F_c t \Bigg\} \end{aligned} \tag{9.1.29}$$

But

$$(-1)^m \cos 2\pi F_c t = \cos 2\pi F_c(t - 2mT)$$

and

$$(-1)^{m+k}\sin 2\pi F_c t = \cos 2\pi F_c(t - 2mT + T)$$

With these substitutions, (9.1.29) reduces to

$$x(t) = \sum_{m=-\infty}^{\infty} x(mT) \frac{\sin(\pi/2T)(t - mT)}{(\pi/2T)(t - mT)} \cos 2\pi F_c(t - mT) \tag{9.1.30}$$

where $T = 1/2B$. This is the desired reconstruction formula for the bandpass signal $x(t)$, with samples taken at the rate of $2B$ samples per second, for the special case in which the upper band frequency $F_c + B/2$ is a multiple of the signal bandwidth B.

In the general case, where only the condition $F_c \geq B/2$ is assumed to hold, let us define the integer part of the ratio $F_c + B/2$ to B as

$$r = \left\lfloor \frac{F_c + B/2}{B} \right\rfloor \tag{9.1.31}$$

While holding the upper cutoff frequency $F_c + B/2$ constant, we increase the bandwidth from B to B' such that

$$\frac{F_c + B/2}{B'} = r \tag{9.1.32}$$

Furthermore, it is convenient to define a new center frequency for the increased bandwidth signal as

$$F_c' = F_c + \frac{B}{2} - \frac{B'}{2} \tag{9.1.33}$$

Clearly, the increased signal bandwidth B' includes the original signal spectrum of bandwidth B.

Now the upper cutoff frequency $F_c + B/2$ is a multiple of B'. Consequently, the signal reconstruction formula in (9.1.30) holds with F_c replaced by F_c' and T replaced by T', where $T' = 1/2B'$, that is,

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT') \frac{\sin(\pi/2T')(t - mT')}{(\pi/2T')(t - mT')} \cos 2\pi F_c'(t - mT') \tag{9.1.34}$$

This proves that $x(t)$ can be represented by samples taken at the uniform rate $1/T' = 2Br'/r$, where r' is the ratio

$$r' = \frac{F_c + B/2}{B} = \frac{F_c}{B} + \frac{1}{2} \tag{9.1.35}$$

and $r = \lfloor r' \rfloor$.

We observe that when the upper cutoff frequency $F_c + B/2$ is not an integer multiple of the bandwidth B, the sampling rate for the bandpass signal must be increased by the factor r'/r. However, note that as F_c/B increases, the ratio r'/r tends toward unity. Consequently, the percent increase in sampling rate tends to zero.

The derivation given above also illustrates the fact that the lowpass signal components $u_c(t)$ and $u_s(t)$ can be expressed in terms of samples of the bandpass signal. Indeed, from (9.1.24), (9.1.25), (9.1.26), and (9.1.27), we obtain the result

$$u_c(t) = \sum_{n=-\infty}^{\infty} (-1)^n x(2nT') \frac{\sin(\pi/2T')(t-2nT')}{(\pi/2T')(t-2nT')} \tag{9.1.36}$$

and

$$u_s(t) = \sum_{n=-\infty}^{\infty} (-1)^{n+r+1} x(2nT'-T') \frac{\sin(\pi/2T')(t-2nT'+T')}{(\pi/2T')(t-2nT'+T')} \tag{9.1.37}$$

where $r = \lfloor r' \rfloor$.

In conclusion, we have demonstrated that a bandpass signal can be represented uniquely by samples taken at a rate

$$2B \leq F_s < 4B$$

where B is the bandwidth of the signal. The lower limit applies when the upper frequency $F_c + B/2$ is a multiple of B. The upper limit on F_s is obtained under worst-case conditions when $r = 1$ and $r' \approx 2$.

9.1.3 Discrete-Time Processing of Continuous-Time Signals

As indicated in our introductory remarks in Chapter 1, there are numerous applications where it is advantageous to process continuous-time (analog) signals on a digital signal processor. Figure 9.5 illustrates the general configuration of the system for digital processing of an analog signal. In designing the processing to be performed, we must first select the bandwidth of the signal to be processed since the signal bandwidth determines the minimum sampling rate. For example, a speech signal, which is to be transmitted digitally, can contain frequency components above 3000 Hz, but for purposes of speech intelligibility and speaker identification, the preservation of frequency components below 3000 Hz is sufficient. Consequently, it would be inefficient from a processing viewpoint to preserve the higher-frequency components and wasteful of channel bandwidth to transmit the extra bits needed to represent these higher-frequency components in the speech signal. Once the desired frequency band is selected we can specify the sampling rate and the characteristics of the prefilter, which is also called an antialiasing filter.

Antialiasing filter. The antialiasing filter is an analog filter which has a twofold purpose. First, it ensures that the bandwidth of the signal to be sampled is limited to the desired frequency range. Thus any frequency components of the signal above the folding frequency $F_s/2$ are sufficiently attenuated so that the

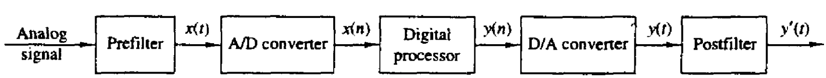


Figure 9.5 Configuration of system for digital processing of an analog signal.

amount of signal distortion due to aliasing is negligible. For example, the speech signal to be transmitted digitally over a telephone channel would be filtered by a lowpass filter having a passband extending to 3000 Hz, a transition band of approximately 400 to 500 Hz, and a stopband above 3400 to 3500 Hz. The speech signal may be sampled at 8000 Hz and hence the folding frequency would be 4000 Hz. Thus aliasing would be negligible.

Another reason for using an antialiasing filter is to limit the additive noise spectrum and other interference, which often corrupts the desired signal. Usually, additive noise is wideband and exceeds the bandwidth of the desired signal. By prefiltering we reduce the additive noise power to that which falls within the bandwidth of the desired signal and we reject the out-of-band noise.

Ideally, we would like to employ a filter with steep cutoff frequency response characteristics and with no delay distortion within the passband. Practically, however, we are constrained to employ filters that have a finite-width transition region, are relatively simple to implement, and introduce some tolerable amount of delay distortion. Very stringent filter specifications, such as a narrow transition region, result in very complex filters. In practice, we may choose to sample the signal well above the Nyquist rate and thus relax the design specifications on the antialiasing filter.

Once we have specified the prefilter requirements and have selected the desired sampling rate, we can proceed with the design of the digital signal processing operations to be performed on the discrete-time signal. The selection of the sampling rate $F_s = 1/T$, where T is the sampling interval, not only determines the highest frequency ($F_s/2$) that is preserved in the analog signal, but also serves as a scale factor that influences the design specifications for digital filters and any other discrete-time systems through which the signal is processed.

For example, suppose that we have an analog signal to be differentiated that has a bandwidth of 3000 Hz. Although differentiation can be performed directly on the analog signal, we choose to do it digitally in discrete time. Hence we sample the signal at the range $F_s = 8000$ Hz and design a digital differentiator as described in Sec. 8.2.4. In this case, the sampling rate $F_s = 8000$ Hz establishes the folding frequency of 4000 Hz, which corresponds to the frequency $\omega = \pi$ in the discrete-time signal. Hence the signal bandwidth of 3000 Hz corresponds to the frequency $\omega_c = 0.75\pi$. Consequently, the discrete-time differentiator for processing the signal would be designed to have a passband of $0 \leq |\omega| \leq 0.75\pi$.

As another example of digital processing, the speech signal that is bandlimited to 3000 Hz and sampled at 8000 Hz may be separated into two or more frequency subbands by digital filtering, and each subband of speech is digitally encoded with different precision, as is done in subband coding (see Section 10.9.5 for more details). The frequency response characteristics of the digital filters for separating the 0- to 3000-Hz signal into subbands are specified relative to the folding frequency of 4000 Hz, which corresponds to the frequency $\omega = \pi$ for the discrete-time signal. Thus we may process any continuous-time signal in the discrete-time domain by performing equivalent operations in discrete time.

The one implicit assumption that we have made in this discussion on the equivalence of continuous-time and discrete-time signal processing is that the quantization error in analog-to-digital conversion and round-off errors in digital signal processing are negligible. These issues are further discussed in this chapter. However, we should emphasize that analog signal processing operations cannot be done very precisely either, since electronic components in analog systems have tolerances and they introduce noise during their operation. In general, a digital system designer has better control of tolerances in a digital signal processing system than an analog system designer who is designing an equivalent analog system.

9.2 ANALOG-TO-DIGITAL CONVERSION

The discussion in Section 9.1 focused on the conversion of continuous-time signals to discrete-time signals using an ideal sampler and ideal interpolation. In this section we deal with the devices for performing these conversions from analog to digital.

Recall that the process of converting a continuous-time (analog) signal to a digital sequence that can be processed by a digital system requires that we quantize the sampled values to a finite number of levels and represent each level by a number of bits. The electronic device that performs this conversion from an analog signal to a digital sequence is called an *analog-to-digital (A/D) converter* (ADC). On the other hand, a *digital-to-analog (D/A) converter* (DAC) takes a digital sequence and produces at its output a voltage or current proportional to the size of the digital word applied to its input. D/A conversion is treated in Section 9.3.

Figure 9.6(a) shows a block diagram of the basic elements of an A/D converter. In this section we consider the performance requirements for these elements. Although we focus mainly on ideal system characteristics, we shall also mention some key imperfections encountered in practical devices and indicate how they affect the performance of the converter. We concentrate on those aspects that are more relevant to signal processing applications. The practical aspects of A/D converters and related circuitry can be found in the manufacturers' specifications and data sheets.

9.2.1 Sample-and-Hold

In practice, the sampling of an analog signal is performed by a sample-and-hold (S/H) circuit. The sampled signal is then quantized and converted to digital form. Usually, the S/H is integrated into the A/D converter.

The S/H is a digitally controlled analog circuit that tracks the analog input signal during the sample mode, and then holds it fixed during the hold mode to the instantaneous value of the signal at the time the system is switched from the

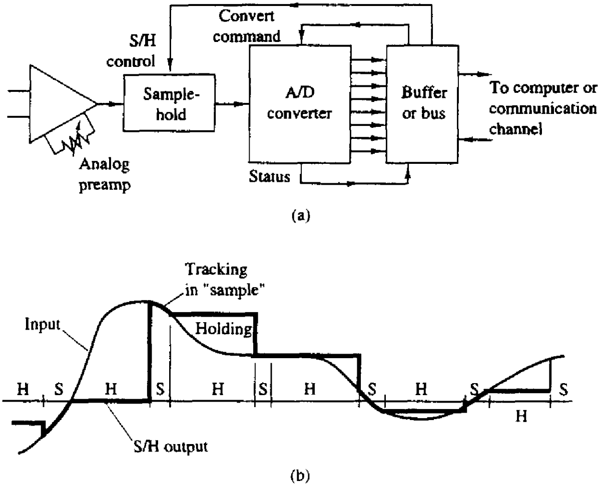


Figure 9.6 (a) Block diagram of basic elements of an A/D converter; (b) time-domain response of an ideal S/H circuit.

sample mode to the hold mode. Figure 9.6(b) shows the time-domain response of an ideal S/H circuit (i.e., a S/H that responds instantaneously and accurately).

The goal of the S/H is to continuously sample the input signal and then to hold that value constant as long as it takes for the A/D converter to obtain its digital representation. The use of an S/H allows the A/D converter to operate more slowly compared to the time actually used to acquire the sample. In the absence of a S/H, the input signal must not change by more than one-half of the quantization step during the conversion, which may be an impractical constraint. Consequently, the S/H is crucial in high-resolution (12 bits per sample or higher) digital conversion of signals that have large bandwidths (i.e., they change very rapidly).

An ideal S/H introduces no distortion in the conversion process and is accurately modeled as an ideal sampler. However, time-related degradations such as errors in the periodicity of the sampling process ("jitter"), nonlinear variations in the duration of the sampling aperture, and changes in the voltage held during conversion ("droop") do occur in practical devices.

The A/D converter begins the conversion after it receives a convert command. The time required to complete the conversion should be less than the duration of the hold mode of the S/H. Furthermore, the sampling period T should be larger than the duration of the sample mode and the hold mode.

In the following sections we assume that the S/H introduces negligible errors and we focus on the digital conversion of the analog samples.

9.2.2 Quantization and Coding

The basic task of the A/D converter is to convert a continuous range of input amplitudes into a discrete set of digital code words. This conversion involves the processes of *quantization* and *coding*. Quantization is a nonlinear and noninvertible process that maps a given amplitude $x(n) \equiv x(nT)$ at time $t = nT$ into an amplitude x_k, taken from a finite set of values. The procedure is illustrated in Fig. 9.7(a), where the signal amplitude range is divided into L intervals

$$I_k = \{x_k < x(n) \leq x_{k+1}\} \qquad k = 1, 2, \ldots, L \tag{9.2.1}$$

by the $L+1$ *decision levels* $x_1, x_L, \ldots, x_{L+1}$. The possible outputs of the quantizer (i.e., the quantization levels) are denoted as $\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_L$. The operation of the quantizer is defined by the relation

$$x_q(n) \equiv Q[x(n)] = \hat{x}_k \qquad \text{if } x(n) \in I_k \tag{9.2.2}$$

In most digital signal processing operations the mapping in (9.2.2) is independent of n (i.e., the quantization is memoryless and is simply denoted as $x_q = Q[x]$). Furthermore, in signal processing we often use *uniform* or *linear quantizers* defined by

$$\begin{aligned} \hat{x}_{k+1} - \hat{x}_k &= \Delta \qquad && k = 1, 2, \ldots, L-1 \\ x_{k+1} - x_k &= \Delta && \text{for finite } x_k, x_{k+1} \end{aligned} \tag{9.2.3}$$

where Δ is the *quantizer step size*. Uniform quantization is usually a requirement if the resulting digital signal is to be processed by a digital system. However, in transmission and storage applications of signals such as speech, nonlinear and time-variant quantizers are frequently used.

If a zero is assigned a quantization level, the quantizer is of the *midtread* type. If zero is assigned a decision level, the quantizer is called a *midrise* type.

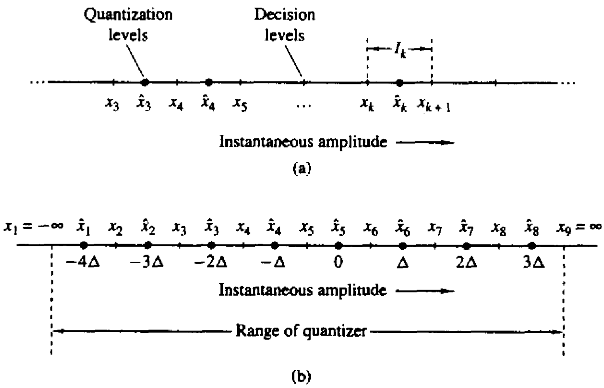


Figure 9.7 Quantization process and an example of a midtread quantizer.

Figure 9.7(b) illustrates a midtread quantizer with $L = 8$ levels. In theory, the extreme decision levels are taken as $x_1 = -\infty$ and $x_{L+1} = \infty$, to cover the total *dynamic range* of the input signal. However, practical A/D converters can handle only a finite range. Hence we define the *range* R of the quantizer by assuming that $I_1 = I_L = \Delta$. For example, the range of the quantizer shown in Fig. 9.7(b) is equal to 8Δ. In practice, the term *full-scale range* (FSR) is used to describe the range of an A/D converter for bipolar signals (i.e., signals with both positive and negative amplitudes). The term *full scale* (FS) is used for unipolar signals.

It can be easily seen that the quantization error $e_q(n)$ is always in the range $-\Delta/2$ to $\Delta/2$:

$$-\frac{\Delta}{2} < e_q(n) \leq \frac{\Delta}{2} \tag{9.2.4}$$

In other words, the instantaneous quantization error cannot exceed half of the quantization step. If the dynamic range of the signal, defined as $x_{\max} - x_{\min}$, is larger than the range of the quantizer, the samples that exceed the quantizer range are clipped, resulting in a large (greater than $\Delta/2$) quantization error.

The operation of the quantizer is better described by the quantization characteristic function, illustrated in Fig. 9.8 for a midtread quantizer with eight

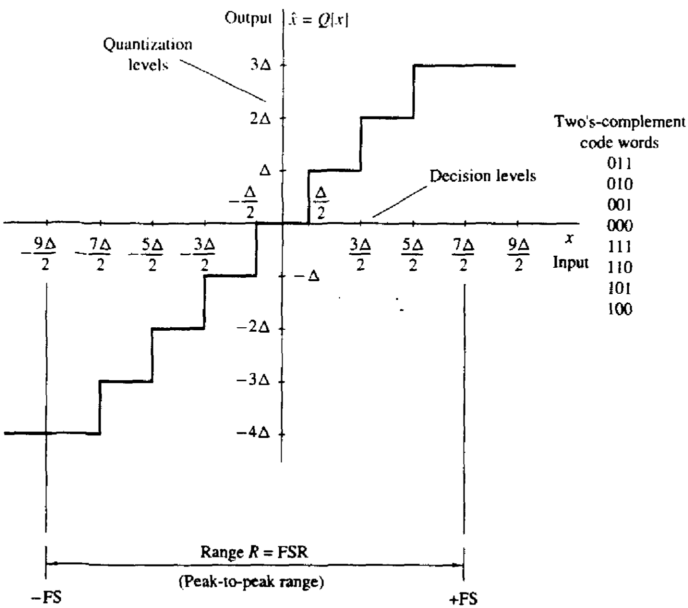


Figure 9.8 Example of a midtread quantizer.

quantization levels. This characteristic is preferred in practice over the midriser because it provides an output that is insensitive to infinitesimal changes of the input signal about zero. Note that the input amplitudes of a midtread quantizer are rounded to the nearest quantization levels.

The *coding* process in an A/D converter assigns a unique binary number to each quantization level. If we have L levels, we need at least L different binary numbers. With a word length of $b+1$ bits we can represent 2^{b+1} distinct binary numbers. Hence we should have $2^{b+1} \geq L$ or, equivalently, $b+1 \geq \log_2 L$. Then the step size or the *resolution* of the A/D converter is given by

$$\Delta = \frac{R}{2^{b+1}} \tag{9.2.5}$$

where R is the range of the quantizer.

There are various binary coding schemes, each with its advantages and disadvantages. Table 9.1 illustrates some existing schemes for 3-bit binary coding. These number representation schemes were described in detail in Section 7.5.

The two's-complement representation is used in most digital signal processors. Thus it is convenient to use the same system to represent digital signals because we can operate on them directly without any extra format conver-

TABLE 9.1 COMMONLY USED BIPOLAR CODES

	Decimal Fraction					
Number	Positive Reference	Negative Reference	Sign + Magnitude	Two's Complement	Offset Binary	One's Complement
+7	$+\frac{7}{8}$	$-\frac{7}{8}$	0 1 1 1	0 1 1 1	1 1 1 1	0 1 1 1
+6	$+\frac{6}{8}$	$-\frac{6}{8}$	0 1 1 0	0 1 1 0	1 1 1 0	0 1 1 0
+5	$+\frac{5}{8}$	$-\frac{5}{8}$	0 1 0 1	0 1 0 1	1 1 0 1	0 1 0 1
+4	$+\frac{4}{8}$	$-\frac{4}{8}$	0 1 0 0	0 1 0 0	1 1 0 0	0 1 0 0
+3	$+\frac{3}{8}$	$-\frac{3}{8}$	0 0 1 1	0 0 1 1	1 0 1 1	0 0 1 1
+2	$+\frac{2}{8}$	$-\frac{2}{8}$	0 0 1 0	0 0 1 0	1 0 1 0	0 0 1 0
+1	$+\frac{1}{8}$	$-\frac{1}{8}$	0 0 0 1	0 0 0 1	1 0 0 1	0 0 0 1
0	0+	0−	0 0 0 0	0 0 0 0	1 0 0 0	0 0 0 0
0	0−	0+	1 0 0 0	(0 0 0 0)	(1 0 0 0)	1 1 1 1
−1	$-\frac{1}{8}$	$+\frac{1}{8}$	1 0 0 1	1 1 1 1	0 1 1 1	1 1 1 0
−2	$-\frac{2}{8}$	$+\frac{2}{8}$	1 0 1 0	1 1 1 0	0 1 1 0	1 1 0 1
−3	$-\frac{3}{8}$	$+\frac{3}{8}$	1 0 1 1	1 1 0 1	0 1 0 1	1 1 0 0
−4	$-\frac{4}{8}$	$+\frac{4}{8}$	1 1 0 0	1 1 0 0	0 1 0 0	1 0 1 1
−5	$-\frac{5}{8}$	$+\frac{5}{8}$	1 1 0 1	1 0 1 1	0 0 1 1	1 0 1 0
−6	$-\frac{6}{8}$	$+\frac{6}{8}$	1 1 1 0	1 0 1 0	0 0 1 0	1 0 0 1
−7	$-\frac{7}{8}$	$+\frac{7}{8}$	1 1 1 1	1 0 0 1	0 0 0 1	1 0 0 0
−8	$-\frac{8}{8}$	$+\frac{8}{8}$		(1 0 0 0)	(0 0 0 0)	

sion. In general, a $(b+1)$-bit binary fraction of the form $\beta_0\beta_1\beta_2\cdots\beta_B$ has the value

$$-\beta_0 \cdot 2^0 + \beta_1 \cdot 2^{-1} + \beta_2 \cdot 2^{-2} + \cdots + \beta_b \cdot 2^{-b}$$

if we use the two's-complement representation. Note that β_0 is the most significant bit (MSB) and β_b is the least significant bit (LSB). Although the binary code used to represent the quantization levels is important for the design of the A/D converter and the subsequent numerical computations, it does not have any effect in the performance of the quantization process. Thus in our subsequent discussions we ignore the process of coding when we analyze the performance of A/D converters.

Figure 9.9(a) shows the characteristic of an ideal 3-bit A/D converter. The only degradation introduced by an ideal converter is the quantization error, which can be reduced by increasing the number of bits. This error, that dominates the performance of practical A/D converters, is analyzed in the next section.

Practical A/D converters differ from ideal converters in several ways. Various degradations are usually encountered in practice. A number of these performance degradations are illustrated in Fig. 9.9(b)–(e). We note that practical A/D converters may have *offset* error (the first transition may not occur at exactly $+\frac{1}{2}$LSB), *scale-factor* (or gain) error (the difference between the values at which the first transition and the last transition occur is not equal to FS − 2LSB), and *linearity* error (the differences between transition values are not all equal or uniformly changing). If the *differential linearity* error is large enough, it is possible for one or more code words to be missed. Performance data on commercially available A/D converters are specified in the manufacturers' data sheets.

9.2.3 Analysis of Quantization Errors

To determine the effects of quantization on the performance of an A/D converter, we adopt a statistical approach. The dependence of the quantization error on the characteristics of the input signal and the nonlinear nature of the quantizer make a deterministic analysis intractable, except in very simple cases.

In the statistical approach, we assume that the quantization error is random in nature. We model this error as noise that is added to the original (unquantized) signal. If the input analog signal is within the range of the quantizer, the quantization error $e_q(n)$ is bounded in magnitude [i.e., $|e_q(n)| < \Delta/2$], and the resulting error is called *granular noise*. When the input falls outside the range of the quantizer (clipping), $e_q(n)$ becomes unbounded and results in *overload noise*. This type of noise can result in severe signal distortion when it occurs. Our only remedy is to scale the input signal so that its dynamic range falls within the range of the quantizer. The following analysis is based on the assumption that there is no overload noise.

The mathematical model for the quantization error $e_q(n)$ is shown in Fig. 9.10. To carry out the analysis, we make the following assumptions about the statistical

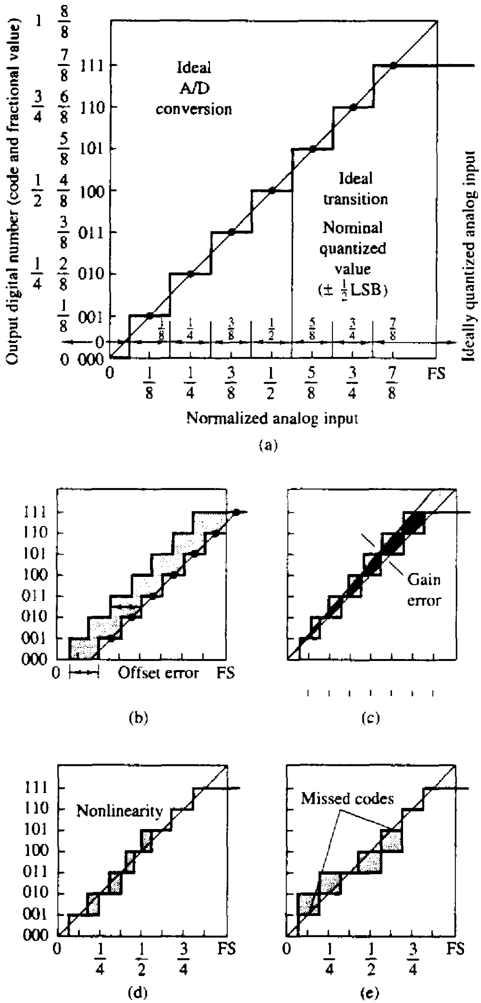


Figure 9.9 Characteristics of ideal and practical A/D converters.

properties of $e_q(n)$:

1. The error $e_q(n)$ is uniformly distributed over the range $-\Delta/2 < e_q(n) < \Delta/2$.
2. The error sequence $\{e_q(n)\}$ is a stationary white noise sequence. In other words, the error $e_q(n)$ and the error $e_q(m)$ for $m \neq n$ are uncorrelated.

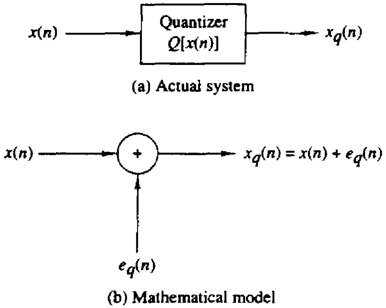


Figure 9.10 Mathematical model of quantization noise.

3. The error sequence $\{e_q(n)\}$ is uncorrelated with the signal sequence $x(n)$.
4. The signal sequence $x(n)$ is zero mean and stationary.

These assumptions do not hold, in general. However, they do hold when the quantization step size is small and the signal sequence $x(n)$ traverses several quantization levels between two successive samples.

Under these assumptions, the effect of the additive noise $e_q(n)$ on the desired signal can be quantified by evaluating the signal-to-quantization noise (power) ratio (SQNR), which can be expressed on a logarithmic scale (in decibels or dB) as

$$\text{SQNR} = 10\log_{10}\frac{P_x}{P_n} \tag{9.2.6}$$

where $P_x = \sigma_x^2 = E[x^2(n)]$ is the signal power and $P_n = \sigma_e^2 = E[e_q^2(n)]$ is the power of the quantization noise.

If the quantization error is uniformly distributed in the range $(-\Delta/2, \Delta/2)$ as shown in Fig. 9.11, the mean value of the error is zero and the variance (the quantization noise power) is

$$P_n = \sigma_e^2 = \int_{-\Delta/2}^{\Delta/2} e^2 p(e)de = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e^2 de = \frac{\Delta^2}{12} \tag{9.2.7}$$

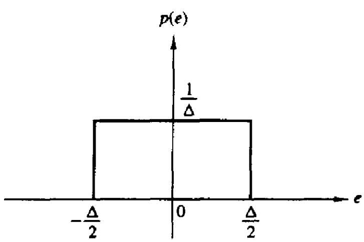


Figure 9.11 Probability density function for the quantization error.

By combining (9.2.5) with (9.2.7) and substituting the result into (9.2.6), the expression for the SQNR becomes

$$\begin{aligned} \text{SQNR} &= 10\log\frac{P_x}{P_n} = 20\log\frac{\sigma_x}{\sigma_e} \\ &= 6.02b + 16.81 - 20\log\frac{R}{\sigma_x} \quad \text{dB} \end{aligned} \tag{9.2.8}$$

The last term in (9.2.8) depends on the range R of the A/D converter and the statistics of the input signal. For example, if we assume that $x(n)$ is Gaussian distributed and the range of the quantizer extends from $-3\sigma_x$ to $3\sigma_x$ (i.e., $R = 6\sigma_x$), then less than 3 out of every 1000 input signal amplitudes would result in an overload on the average. For $R = 6\sigma_x$, (9.2.8) becomes

$$\text{SQNR} = 6.02b + 1.25 \text{ dB}$$

The formula in (9.2.8) is frequently used to specify the precision needed in an A/D converter. It simply means that each additional bit in the quantizer increases the signal-to-quantization noise ratio by 6 dB. (It is interesting to note that the same result was derived in Section 1.4 for a sinusoidal signal using a deterministic approach.) However, we should bear in mind the conditions under which this result has been derived.

Due to limitations in the fabrication of A/D converters, their performance falls short of the theoretical value given by (9.2.8). As a result, the effective number of bits may be somewhat less than the number of bits in the A/D converter. For instance, a 16-bit converter may have only an effective 14 bits of accuracy.

9.2.4 Oversampling A/D Converters

The basic idea in oversampling A/D converters is to increase the sampling rate of the signal to the point where a low-resolution quantizer suffices. By oversampling, we can reduce the dynamic range of the signal values between successive samples and thus reduce the resolution requirements on the quantizer. As we have observed in the preceding section, the variance of the quantization error in A/D conversion is $\sigma_e^2 = \Delta^2/12$, where $\Delta = R/2^{b+1}$. Since the dynamic range of the signal, which is proportional to its standard deviation σ_x, should match the range R of the quantizer, it follows that Δ is proportional to σ_x. Hence for a given number of bits, the power of the quantization noise is proportional to the variance of the signal to be quantized. Consequently, for a given fixed SQNR, a reduction in the variance of the signal to be quantized allows us to reduce the number of bits in the quantizer.

The basic idea for reducing the dynamic range leads us to consider *differential quantization*. To illustrate this point, let us evaluate the variance of the difference between two successive signal samples. Thus we have

$$d(n) = x(n) - x(n-1) \tag{9.2.9}$$

The variance of $d(n)$ is

$$\begin{aligned}\sigma_d^2 &= E[d^2(n)] = E\{[x(n) - x(n-1)]^2\}\\ &= E[x^2(n)] - 2E[x(n)x(n-1)] + E[x^2(n-1)]\\ &= 2\sigma_x^2[1 - \gamma_{xx}(1)]\end{aligned} \tag{9.2.10}$$

where $\gamma_{xx}(1)$ is the value of the autocorrelation sequence $\gamma_{xx}(m)$ of $x(n)$ evaluated at $m = 1$. If $\gamma_{xx}(1) > 0.5$, we observe that $\sigma_d^2 < \sigma_x^2$. Under this condition, it is better to quantize the difference $d(n)$ and to recover $x(n)$ from the quantized values $\{d_q(n)\}$. To obtain a high correlation between successive samples of the signal, we require that the sampling rate be significantly higher than the Nyquist rate.

An even better approach is to quantize the difference

$$d(n) = x(n) - ax(n-1) \tag{9.2.11}$$

where a is a parameter selected to minimize the variance in $d(n)$. This leads to the result (see Problem 9.7) that the optimum choice of a is

$$a = \frac{\gamma_{xx}(1)}{\gamma_{xx}(0)} = \frac{\gamma_{xx}(1)}{\sigma_x^2}$$

and

$$\sigma_d^2 = \sigma_x^2[1 - a^2] \tag{9.2.12}$$

In this case, $\sigma_d^2 < \sigma_x^2$, since $0 \le a \le 1$. The quantity $ax(n-1)$ is called a first-order predictor of $x(n)$.

Figure 9.12 shows a more general *differential predictive* signal quantizer system. This system is used in speech encoding and transmission over telephone channels and is known as differential pulse code modulation (DPCM). The goal of the predictor is to provide an estimate $\hat{x}(n)$ of $x(n)$ from a linear combination of past values of $x(n)$, so as to reduce the dynamic range of the difference signal $d(n) = x(n) - \hat{x}(n)$. Thus a predictor of order p has the form

$$\hat{x}(n) = \sum_{k=1}^{p} a_k x(n-k) \tag{9.2.13}$$

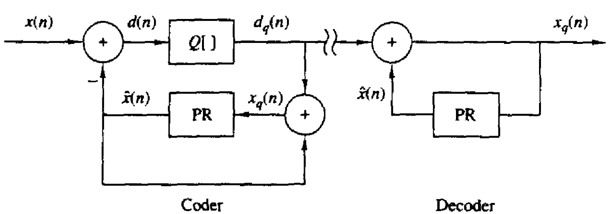


Figure 9.12 Encoder and decoder for differential predictive signal quantizer system.

The use of the feedback loop around the quantizer as shown in Fig. 9.12 is necessary to avoid the accumulation of quantization errors at the decoder. In this configuration, the error $e(n) = d(n) - d_q(n)$ is

$$e(n) = d(n) - d_q(n) = x(n) - \hat{x}(n) - d_q(n) = x(n) - x_q(n)$$

Thus the error in the reconstructed quantized signal $x_q(n)$ is equal to the quantization error for the sample $d(n)$. The decoder for DPCM that reconstructs the signal from the quantized values is also shown in Fig. 9.12.

The simplest form of differential predictive quantization is called *delta modulation* (DM). In DM, the quantizer is a simple 1-bit (two-level) quantizer and the predictor is a first-order predictor, as shown in Fig. 9.13(a). Basically, DM provides a staircase approximation of the input signal. At every sampling instant, the sign of the difference between the input sample $x(n)$ and its most recent staircase approximation $\hat{x}(n) = ax_q(n-1)$ is determined, and then the staircase signal is updated by a step Δ in the direction of the difference.

From Fig. 9.13(a) we observe that

$$x_q(n) = ax_q(n-1) + d_q(n) \tag{9.2.14}$$

which is the discrete-time equivalent of an analog integrator. If $a = 1$, we have an ideal accumulator (integrator) whereas the choice $a < 1$ results in a "leaky integrator." Figure 9.13(c) shows an analog model that illustrates the basic principle for the practical implementation of a DM system. The analog lowpass filter is necessary for the rejection of out-of-band components in the frequency range between B and $F_s/2$, since $F_s >> B$ due to oversampling.

The crosshatched areas in Fig. 9.13(b) illustrate two types of quantization error in DM, slope-overload distortion and granular noise. Since the maximum slope Δ/T in $x(n)$ is limited by the step size, slope-overload distortion can be avoided if max $|dx(t)/dt| \leq \Delta/T$. The granular noise occurs when the DM tracks a relatively flat (slowly changing) input signal. We note that increasing Δ reduces overload distortion but increases the granular noise, and vice versa.

One way to reduce these two types of distortion is to use an integrator in front of the DM, as shown in Fig. 9.14(a). This has two effects. First, it emphasizes the low frequencies of $x(t)$ and increases the correlation of the signal into the DM input. Second, it simplifies the DM decoder because the differentiator (inverse system) required at the decoder is canceled by the DM integrator. Hence the decoder is simply a lowpass filter, as shown in Fig. 9.14(a). Furthermore, the two integrators at the encoder can be replaced by a single integrator placed before the comparator, as shown in Fig. 9.14(b). This system is known as *sigma-delta modulation* (SDM).

SDM is an ideal candidate for A/D conversion. Such a converter takes advantage of the high sampling rate and spreads the quantization noise across the band up to $F_s/2$. Since $F_s >> B$, the noise in the signal-free band $B \leq F \leq F_s/2$ can be removed by appropriate digital filtering. To illustrate this principle, let

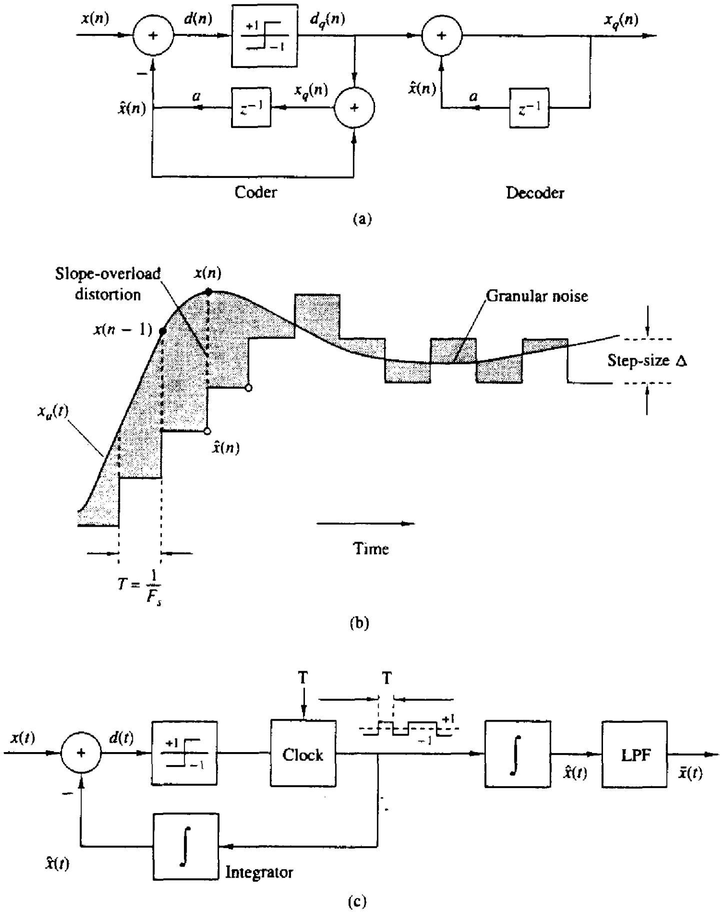
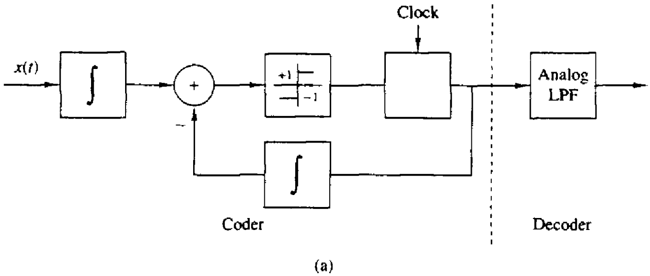


Figure 9.13 Delta modulation system and two types of quantization errors.

us consider the discrete-time model of SDM, shown in Fig. 9.15, where we have assumed that the comparator (1-bit quantizer) is modeled by an additive white noise source with variance $\sigma_e^2 = \Delta^2/12$. The integrator is modeled by the discrete-time system with system function

$$H(z) = \frac{z^{-1}}{1 - z^{-1}} \tag{9.2.15}$$



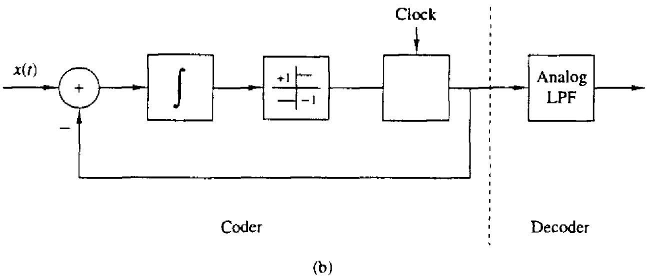


Figure 9.14 Sigma-delta modulation system.

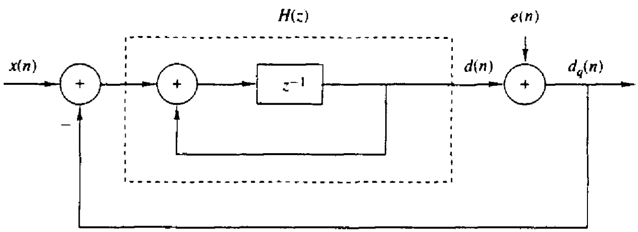


Figure 9.15 Discrete-time model of sigma-delta modulation.

The z-transform of the sequence $\{d_q(n)\}$ is

$$\begin{aligned} D_q(z) &= \frac{H(z)}{1+H(z)}X(z) + \frac{1}{1+H(z)}E(z) \\ &= H_s(z)X(z) + H_n(z)E(z) \end{aligned} \tag{9.2.16}$$

where $H_s(z)$ and $H_n(z)$ are the signal and noise system functions, respectively. A good SDM system has a flat frequency response $H_s(\omega)$ in the signal frequency band $0 \leq F \leq B$. On the other hand, $H_n(z)$ should have high attenuation in the frequency band $0 \leq F \leq B$ and low attenuation in the band $B \leq F \leq F_s/2$.

For the first-order SDM system with the integrator specified by (9.2.15), we have

$$H_s(z) = z^{-1} \qquad H_n(z) = 1 - z^{-1} \tag{9.2.17}$$

Thus $H_s(z)$ does not distort the signal. The performance of the SDM system is therefore determined by the noise system function $H_n(z)$, which has a magnitude frequency response

$$|H_n(F)| = 2\left|\sin\frac{\pi F}{F_s}\right| \tag{9.2.18}$$

as shown in Fig. 9.16. The in-band quantization noise variance is given as

$$\sigma_n^2 = \int_{-B}^{B} |H_n(F)|^2 S_e(F)\,dF \tag{9.2.19}$$

where $S_e(F) = \sigma_e^2/F_s$ is the power spectral density of the quantization noise. From this relationship we note that doubling F_s (increasing the sampling rate by a factor of 2), while keeping B fixed, reduces the power of the quantization noise by 3 dB. This result is true for any quantizer. However, additional reduction may be possible by properly choosing the filter $H(z)$.

For the first-order SDM, it can be shown (see Problem 9.10) that for $F_s >> 2B$, the in-band quantization noise power is

$$\sigma_n^2 \approx \frac{1}{3}\pi^2\sigma_e^2\left(\frac{2B}{F_s}\right)^3 \tag{9.2.20}$$

Note that doubling the sampling frequency reduces the noise power by 9 dB of which 3 dB is due to the reduction in $S_e(F)$ and 6 dB is due to the filter characteristic $H_n(F)$. An additional 6-dB reduction can be achieved by using a double integrator (see Problem 9.11).

In summary, the noise power σ_n^2 can be reduced by increasing the sampling rate to spread the quantization noise power over a larger frequency band $(-F_s/2, F_s/2)$, and then shaping the noise power spectral density by means of an

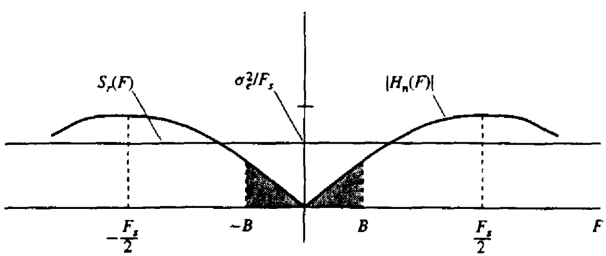


Figure 9.16 Frequency (magnitude) response of noise system function.

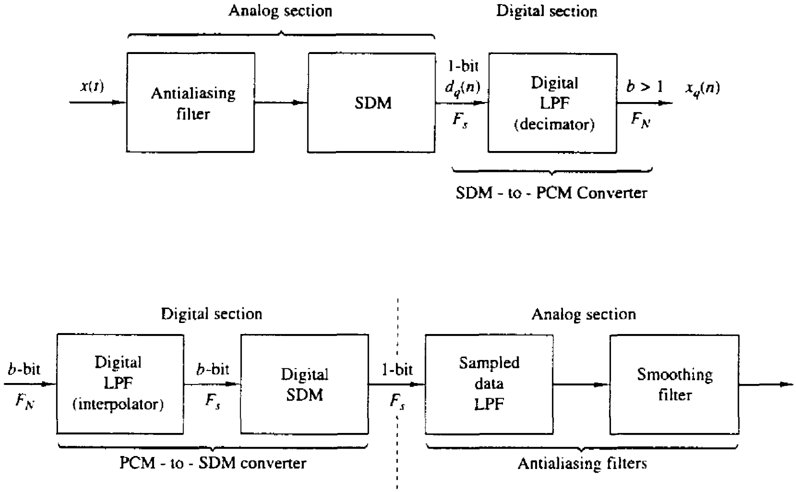


Figure 9.17 Basic elements of an oversampling A/D converter.

appropriate filter. Thus, SDM provides a 1-bit quantized signal at a sampling frequency $F_s = 2IB$, where the oversampling (interpolation) factor I determines the SNR of the SDM quantizer.

Next, we explain how to convert this signal into a b-bit quantized signal at the Nyquist rate. First, we recall that the SDM decoder is an analog lowpass filter with a cutoff frequency B. The output of this filter is an approximation to the input signal $x(t)$. Given the 1-bit signal $d_q(n)$ at sampling frequency F_s, we can obtain a signal $x_q(n)$ at a lower sampling frequency, say the Nyquist rate of $2B$ or somewhat faster, by resampling the output of the lowpass filter at the $2B$ rate. To avoid aliasing, we first filter out the out-of-band $(B, F_s/2)$ noise by processing the wideband signal. The signal is then passed through the lowpass filter and resampled (downsampled) at the lower rate. The downsampling process is called *decimation* and is treated in great detail in Chapter 10.

For example, if the interpolation factor is $I = 256$, the A/D converter output can be obtained by averaging successive non-overlapping blocks of 128 bits. This averaging would result in a digital signal with a range of values from zero to $256 (b \approx 8$ bits) at the Nyquist rate. The averaging process also provides the required antialiasing filtering.

Figure 9.17 illustrates the basic elements of an oversampling A/D converter. Oversampling A/D converters for voice-band (3-kHz) signals are currently fabricated as integrated circuits. Typically, they operate at a 2-MHz sampling rate, downsample to 8 kHz, and provide 16-bit accuracy.

9.3 DIGITAL-TO-ANALOG CONVERSION

In Section 4.2.9 we demonstrated that a bandlimited lowpass analog signal, which has been sampled at the Nyquist rate (or faster), can be reconstructed from its samples without distortion. The ideal reconstruction formula or ideal interpolation formula derived in Section 4.2.9 is

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\frac{\sin(\pi/T)(t-nT)}{(\pi/T)(t-nT)} \tag{9.3.1}$$

where the sampling interval $T = 1/F_s = 1/2B$, F_s is the sampling frequency and B is the bandwidth of the analog signal.

We have viewed the reconstruction of the signal $x(t)$ from its samples as an interpolation problem and have described the function

$$g(t) = \frac{\sin(\pi t/T)}{\pi t/T} \tag{9.3.2}$$

as the ideal interpolation function. The interpolation formula for $x(t)$, given by (9.3.1), is basically a linear superposition of time-shifted versions of $g(t)$, with each $g(t-nT)$ weighted by the corresponding signal sample $x(nT)$.

Alternatively, we can view the reconstruction of the signal from its samples as a linear filtering process in which a discrete-time sequence of short pulses (ideally impulses) with amplitudes equal to the signal samples, excites an analog filter, as illustrated in Fig. 9.18. The analog filter corresponding to the ideal interpolator has a frequency response

$$H(F) = \begin{cases} T, & |F| \le \dfrac{1}{2T} = \dfrac{F_s}{2} \\ 0, & |F| > \dfrac{1}{2T} \end{cases} \tag{9.3.3}$$

$H(F)$ is simply the Fourier transform of the interpolation function $g(t)$. In other words, $H(F)$ is the frequency response of an analog reconstruction filter whose

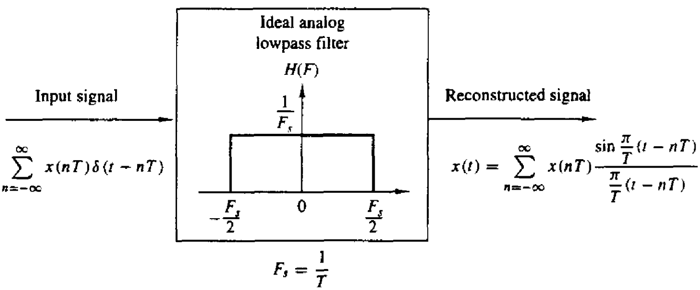


Figure 9.18 Signal reconstruction viewed as a filtering process.